

Renal Handling of Organic Solutes

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CHAPTER OUTLINE

GLUCOSE, 218

AMINO ACIDS, 239

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KEY POINTS

- In a normoglycemic adult, the kidneys filter 160–180 g/day of glucose (~30% of daily energy expenditure), which is reabsorbed by the sodium glucose cotransporter SGLT2 (~97%) and SGLT1 (~3%) in the early and late proximal tubule, respectively.
- The basal overall glucose tubular reabsorption capacities for SGLT2 versus SGLT1 is in the range of 3:1 to 5:1. The transport capacity of tubular SGLT1 is unmasked (up to ~80 g/day) when more glucose is delivered to the late proximal tubule (e.g., in diabetes or with SGLT2 inhibition).
- The good correlation between the level of basolateral GLUT1 expression and the glycolytic activity of the different nephron segments indicates that the more distal tubule segments in particular are taking up glucose for energy supply via basolateral GLUT1.
- One of the major roles of organic anionic transporters, long considered mainly drug and toxin transporters, now appears to be to regulate many aspects of endogenous physiology.
- OAT1 and OAT3 appear to be the major renal basolateral transporters involved in the elimination of numerous uremic toxins originating in the gut microbiome, although OCT2 is likely the main route of TMAO elimination.
- The Remote Sensing and Signaling Hypothesis is a systems biology theory about the role of SLC and ABC transporters in the interorgan and interorganismal (e.g., host gut microbiome) “remote” communication via transporter-mediated movement of metabolites, signaling molecules, gut microbiome products, nutrients, uric acid, and uremic toxins into different body tissues and fluid compartments. It provides a framework for understanding uremia and hyperuricemia.
- Amino acid transporters often form heterodimers with ancillary subunits that are essential for trafficking of the transporters to the cell surface. Genetic complexity is observed in renal aminoacidurias due to heterodimer formation and transporter redundancy.
- Rare inherited aminoacidurias define four major routes of amino acid reabsorption. Amino acid antiporters play an important role in the apical and basolateral transport of cationic and neutral amino acids.

GLUCOSE

The kidneys are a major site of glucose handling. This includes the continuous glomerular filtration of large amounts of glucose, almost all of which is subsequently reabsorbed by the proximal tubule, such that the formed urine in a healthy individual is nearly glucose free. The glucose reabsorbed by

the proximal tubule is primarily taken up into the peritubular capillaries and provided as an energy source to further distal tubular segments or returned to the systemic circulation. Moreover and in addition to the liver, the kidneys contribute to the endogenous production of glucose or gluconeogenesis. Thus, the kidneys use glucose as fuel but also contribute to maintaining blood glucose levels and overall metabolic balance by reabsorbing filtered glucose and generating new glucose.

This is relevant in healthy individuals, in particular in the fasting state, and becomes pathophysiologically important in diabetes and hyperglycemic conditions. As a consequence, new antihyperglycemic drugs have been developed that target renal glucose reabsorption, induce urinary glucose loss, and have clinical efficacy with regard to lowering blood glucose, and even more importantly, they have protective effects on the kidney and cardiovascular system.

PHYSIOLOGY OF RENAL GLUCOSE TRANSPORT

In many organisms including human, the cellular uptake and metabolism of D-glucose serves as an important energy source.^{1,2} The brain primarily runs on glucose and depends on its continuous uptake, which alone requires ~125 g of glucose every day. As a consequence, glucose homeostasis is finely regulated, and blood glucose levels are maintained in a range of 4–9 mmol/L by various hormones, including insulin and glucagon, that regulate glucose uptake into target cells as well as glucose storage and endogenous glucose production.^{1,2}

Glucose is a small molecule that is freely filtered by the glomeruli of the kidneys. Under conditions of normal blood glucose levels (~5.5 mmol/L or 100 mg/dL) and normal glomerular filtration rate (GFR, ~180 L/day), the kidneys filter 160–180 g of glucose each day. This is equal to ~30% of the daily energy expenditure, which would be lost into the urine if not regained by the renal tubules. Instead, more than 99% of the filtered glucose is reabsorbed by the tubules, primarily in the proximal tubule (Fig. 8.1). As described in

more detail below, glucose reabsorption in the proximal tubule involves the two Na⁺-glucose cotransporters SGLT2 and SGLT1, which are expressed in the brush border membrane of the early and later proximal tubule, respectively. Na⁺-glucose cotransport is a saturable process and has a maximum transport capacity (T_{max}). The T_{max} of the kidneys for glucose can vary among individuals and averages ~430 g and ~500 g/day (300 and 350 mg/min) in female and male healthy human subjects, respectively.^{3,4} This equals ~threefold the normal tubular glucose load of 160–180 g/day so that the renal glucose reabsorption capacity is not saturated under conditions of normal blood glucose levels and GFR. Theoretically, at a normal GFR, the T_{max} should be reached, and glucose should begin to be excreted in the urine at a plasma glucose threshold of ~15.5 mmol/L (280 mg/dL). The T_{max} for glucose of individual nephrons is variable, however, and thus, low-level urinary glucose spilling begins in a normal, glucose-tolerant individual at modestly elevated plasma glucose levels of ~10–11 mmol/L (180–200 mg/dL; see “Splay,” Fig. 8.2). A robust and linear increase in urinary glucose excretion occurs when blood glucose levels rise above 15–16 mmol/L. GFR is a determinant of the filtered glucose load, and as a consequence, glucosuria can occur at lower plasma glucose concentrations when GFR is elevated (e.g., in pregnancy or in diabetes), or at higher blood glucose levels when GFR is reduced (e.g., in chronic kidney disease; CKD). In addition, changes in the transport activity and expression level of SGLT2 and SGLT1 (see later) are expected to further modify this relationship.

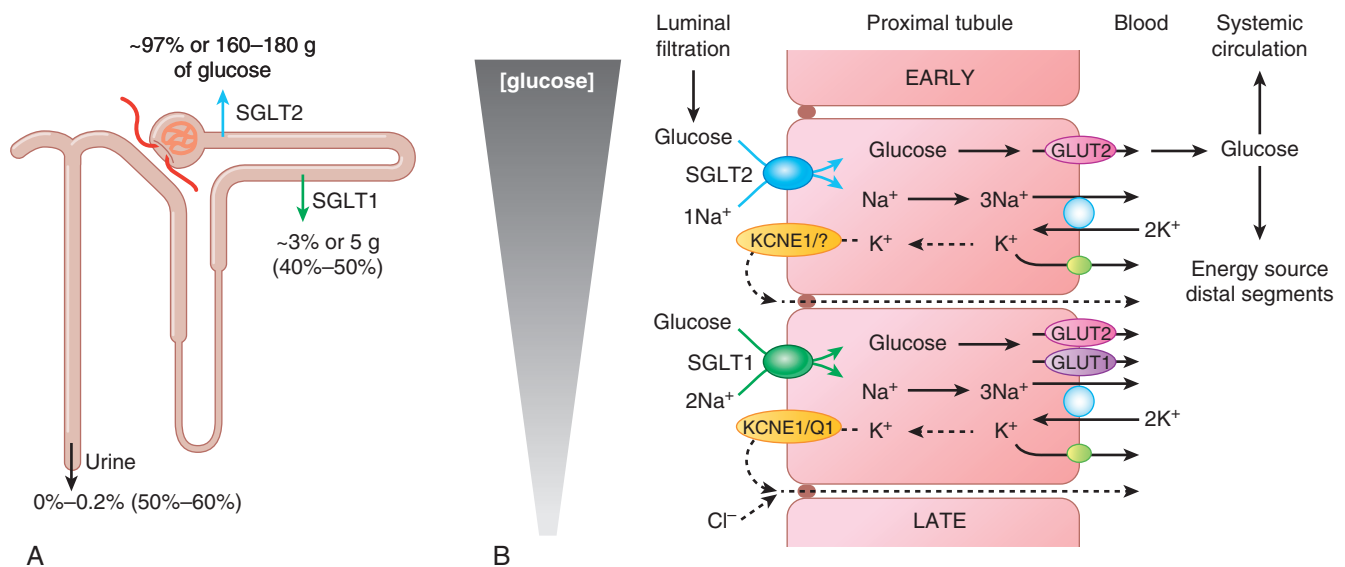


Fig. 8.1 Glucose transport in the kidney. (A) Under normoglycemia, ~97% of filtered glucose is reabsorbed via SGLT2 in the early segments of the proximal tubule. The remaining ~3% of glucose is reabsorbed by SGLT1 in the late proximal tubule such that the urine is nearly free of glucose. SGLT2 inhibition shifts glucose reabsorption downstream and unmasks the capacity of SGLT1 to reabsorb glucose (~40% of filtered glucose, depending on glucose load; see numbers in parentheses). (B) Cell model of glucose transport: The basolateral Na⁺-K⁺-ATPase lowers cytosolic Na⁺ concentrations and generates a negative interior voltage, thereby providing the driving force for Na⁺-coupled glucose uptake through SGLT2 and SGLT1 across the apical membrane. The facilitative glucose transporters GLUT2 and GLUT1 mediate glucose transport across the basolateral membrane down its chemical gradient. Na⁺-glucose cotransport is electrogenic and accompanied by paracellular Cl⁻ reabsorption or transcellular K⁺ secretion to stabilize membrane potential; K⁺ channels KCNE1/unknown α subunit and KCNE1/KCNQ1 in early and late proximal tubule, respectively. (This figure was modified with permission from Vallon V. Molecular determinants of renal glucose transport. *Am J Physiol Cell Physiol.* 2011;300:C6–C8.)

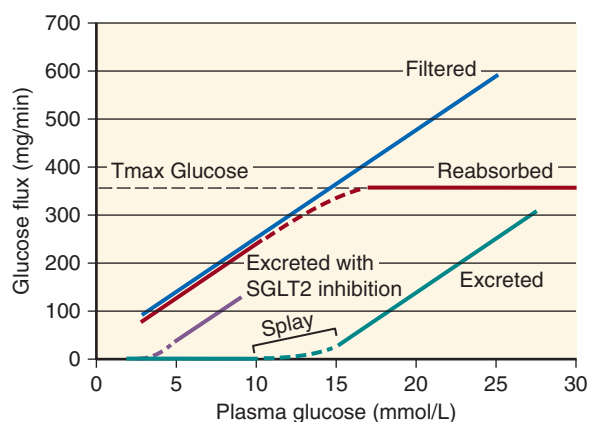


Fig. 8.2 Tubular reabsorption and urinary excretion of glucose as a function of filtered glucose. Tubular reabsorption of glucose increases linearly with the filtered glucose load until reabsorption reaches the tubular reabsorption capacity (*T_{max} Glucose*) and glucose starts appearing in the urine. Theoretically, a *T_{max}* of ~350 mg/min and normal glomerular filtration rate would result in a plasma glucose threshold of ~15.5 mmol/L. However, due to variability of *T_{max}* in individual nephrons, the plasma glucose concentration that results in glucosuria in a normal individual is ~10–11.1 mmol/L (see “*Splay*”). SGLT2 inhibition reduces the renal glucose reabsorption to the transport capacity of SGLT1 and shifts the function to the left—that is, it reduces the renal glucose threshold (~3 mM) and *T_{max}* (~150 mg/min).

A PRIMARY ROLE FOR SGLT2 IN RENAL GLUCOSE REABSORPTION

Experiments performed on isolated nephron segments of rabbit kidneys in the early 1980s identified differences in the rate of uptake and affinity for glucose between the early and late proximal tubule segments, respectively.⁵ Subsequent studies confirmed that the heterogeneity in glucose transport across the proximal tubule was attributed to the presence of two different glucose transporters in the brush border membrane.⁶ These studies and transport studies in membrane vesicles and analyses of mRNA expression in isolated nephron segments of rat and rabbit kidneys as well as the cloning of the responsible genes, performed largely between 1981 and 1995, identified the Na⁺–glucose cotransporters SGLT2 (SLC5A2) and SGLT1 (SLC5A1) as the primary genes and pathways for renal glucose reabsorption.^{5–14} These studies established the concept that the “bulk” of tubular glucose uptake across the apical membrane occurs in the “early” proximal tubule (S1/2 segment) and is mediated by the low-affinity and high-capacity SGLT2. In comparison, the higher-affinity and lower-capacity SGLT1 is thought to “clean up” most of the remaining luminal glucose in “later” parts of the proximal tubule (S2/S3 segment) (see Fig. 8.1). In accordance, SGLT2 and SGLT1 have been localized with the use of well-validated antibodies in both rodents and humans to the brush border membrane primarily of the early and late proximal tubule, respectively.^{15–18} In the mouse kidney, the levels of SGLT1 protein expression in the brush border were highest in S2 segments and somewhat lower in S3 segments in medullary rays and in the outer stripe.¹⁹ In the human kidney, the strongest expression of SGLT1 was found in the S3 segment.¹⁸ In accordance and demonstrating the functional role of SGLT2, free-flow renal micropuncture

showed that glucose reabsorption in the early proximal tubule is completely absent in mice lacking SGLT2¹⁵ (Fig. 8.3). In comparison, fractional glucose reabsorption along the proximal convoluted tubule (PCT) was only slightly reduced from 97% to 94% in mice lacking SGLT1.²⁰

The phenotype of humans carrying mutations in the genes for SGLT1 (*SLC5A1*) and SGLT2 (*SLC5A2*) demonstrated their distinct quantitative contribution to renal glucose reabsorption. Mutations in SGLT1 cause “Intestinal Glucose Galactose Malabsorption” [Online Mendelian Inheritance in Man (OMIM) 182380] because the active intestinal reabsorption of glucose is mediated by SGLT1.^{21,22} Newborns with mutations in SGLT1 as well as mice lacking SGLT1²⁰ can present with life-threatening diarrhea when exposed to dietary galactose or glucose; however, they show little or no glucosuria. In contrast, individuals with mutations in SGLT2 present persistent “Familial Renal Glucosuria” (OMIM 233100) ranging from 1 to >100 g per day, but they have no intestinal phenotype.²³ Although mutations in SGLT2 are rare and therefore the consequences are not well studied or fully understood, it is remarkable that no other complications (e.g., urinary infections or impaired kidney function) have been consistently observed in these subjects.^{21,23} This information added to the rationale of developing SGLT2 inhibitors as potentially safe antihyperglycemic drugs (see later). Consistent with the described phenotypes in humans, genetic and pharmacologic studies in mice showed that under normoglycemic conditions, SGLT2 reabsorbs ~97% of the filtered glucose, whereas SGLT1 mediates the reabsorption of the remaining ~2%–3%^{15,20,24} (see Figs. 8.1 and 8.3).

UNMASKING A SIGNIFICANT GLUCOSE TRANSPORT CAPACITY OF SGLT1 IN THE LATE PROXIMAL TUBULE

In healthy human subjects, similar to the phenotype in rodents, SGLT2 is thought to reabsorb >90% of filtered glucose, yet they maintain a fractional glucose reabsorption of 40%–50% following application of a selective SGLT2 inhibitor^{25–27} (see Fig. 8.1). This observation is mimicking the phenotype of normoglycemic mice lacking *Sgt2* (*Sgt2*^{−/−}), in which fractional renal glucose reabsorption varied between 10% and 60%, inversely with the amount of filtered glucose, with a mean value of ~40%¹⁵ (see Fig. 8.3). Follow-up studies demonstrated that the persisting glucose reabsorption is mediated in the downstream late proximal tubule by SGLT1, whose transport capacity is unmasked by SGLT2 inhibition (Figs. 8.1 and 8.3). First and indirect evidence came from micropuncture studies in *Sgt2*^{−/−} mice: these mice have no net glucose reabsorption in the early proximal tubule; however, their glucose reabsorption in the later parts of the PCT accessible to micropuncture, where SGLT1 is expressed in S2 segments, is increased compared with wild-type (WT) mice¹⁵ (Fig. 8.3). Metabolic cage studies further showed that the dose–response curve for glucosuria of a selective SGLT2 inhibitor was shifted leftward in *Sgt1*^{−/−} mice; that is, glucosuria initiated at lower doses, and the maximum glucosuric response doubled compared with WT mice²⁴ (Fig. 8.3). Renal clearance studies found that concentrations of the SGLT2 inhibitor in early proximal tubule fluid close to the reported half maximal inhibitory concentration (*IC*₅₀) for mouse SGLT2 were associated with fractional renal glucose reabsorption

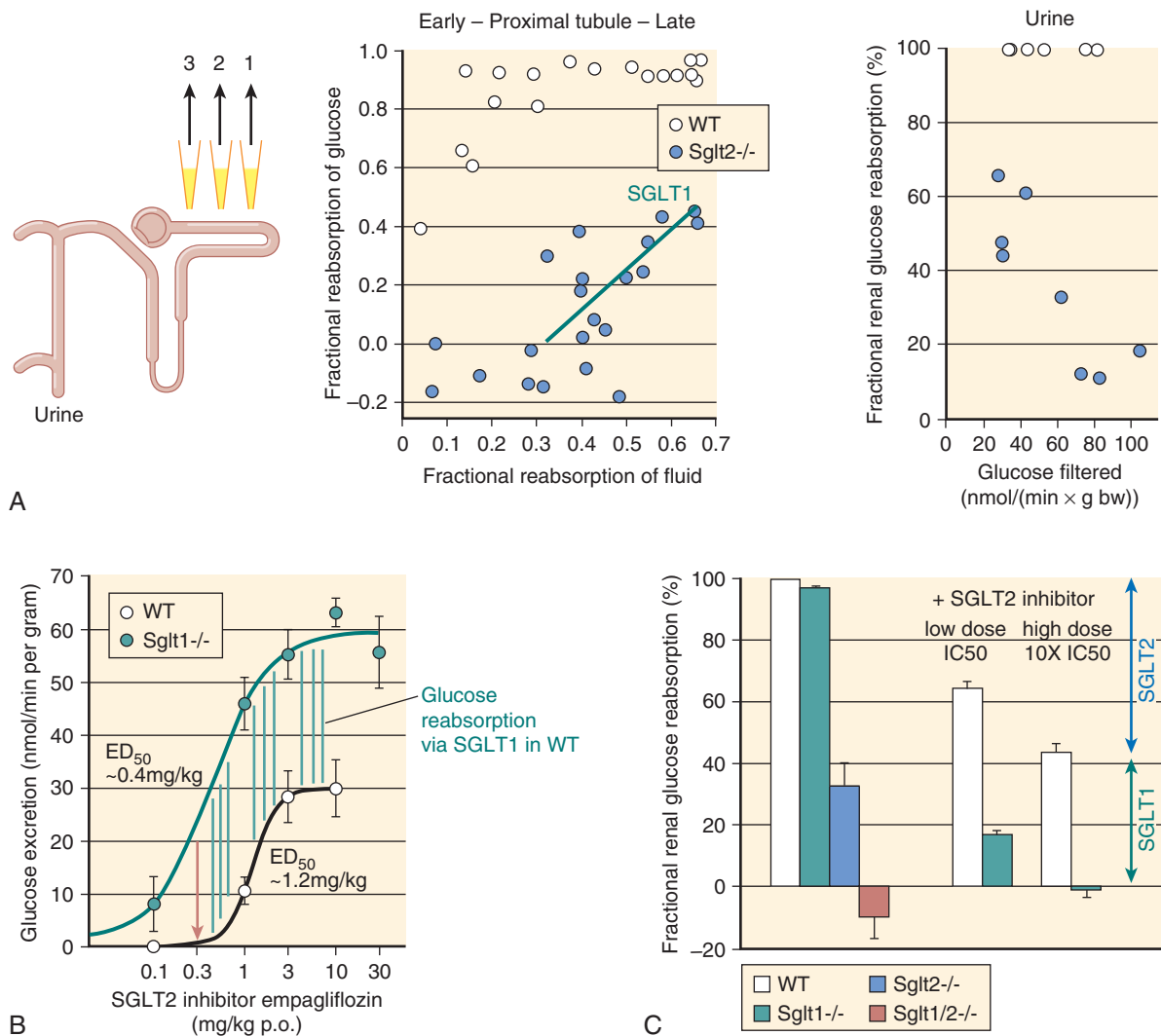


Fig. 8.3 Contribution of SGLT2 and SGLT1 to renal glucose reabsorption. (A) *Left panels:* Free-flow collections of tubular fluid performed by micropuncture to establish a profile for fractional reabsorption of glucose versus fluid along accessible proximal tubules at the kidney surface. Glucose reabsorption is absent in the early proximal tubule of *Sglt2*^{-/-} mice but enhanced in the later proximal tubule, potentially reflecting compensation by SGLT1-mediated transport. *Right panel:* Renal inulin clearance studies revealed that the reduction in fractional renal glucose reabsorption in *Sglt2*^{-/-} mice was inversely related to the amount of filtered glucose. (B) In metabolic cage studies, the SGLT2 inhibitor empagliflozin dose dependently increased glucose excretion in wild-type (WT) mice. The response curve was shifted leftward, and the maximum response doubled in *Sglt1*^{-/-} mice. The difference between dose-response curves, which reflects glucose reabsorption via SGLT1 in WT mice, was maintained for higher doses (all vertical lines have same length), consistent with selectivity of the inhibitor for SGLT2 versus SGLT1 in this dose range. Glucosuria is initiated in WT mice when SGLT1-mediated glucose uptake is saturate (red arrow). (C) Renal inulin clearance studies in mice lacking *Sglt1*, *Sglt2*, or both *Sglt1* and *Sglt2* indicated that the glucose reabsorption preserved in *Sglt2*^{-/-} (~40%) is mediated by SGLT1. Application of the SGLT2 inhibitor empagliflozin at low and high doses to establish free plasma concentrations (corresponding to early tubular concentrations) close to IC₅₀ for mouse SGLT2 (~1–2 nM) or 10-fold this concentration confirmed the role of SGLT1 during pharmacologic SGLT2 inhibition. (Data from Vallon V, Platt KA, Cunard R, et al. SGLT2 mediates glucose reabsorption in the early proximal tubule. *J Am Soc Nephrol.* 2011;22:104–112; and Rieg T, Masuda T, Gerasimova M, et al. Increase in SGLT1-mediated transport explains renal glucose reabsorption during genetic and pharmacological SGLT2 inhibition in euglycemia. *Am J Physiol Renal Physiol.* 2014;306:F188–F193.)

of 64% in WT and 17% in *Sglt1*^{-/-} mice. Dosing the SGLT2 inhibitor to fully inhibit SGLT2 reduced fractional renal glucose reabsorption to 44% in WT and eliminated net renal glucose reabsorption in *Sglt1*^{-/-} mice (Fig. 8.3). Finally, the absence of net renal glucose reabsorption was confirmed in male and female mice lacking both *Sglt1* and *Sglt2*²⁴ (Fig. 8.3). These studies demonstrated that SGLT1 provides a significant glucose transport capacity in the late proximal tubule, which, in the normal kidney, remains mostly unused

due to upstream glucose reabsorption by SGLT2. Inhibition of the latter, however, delivers more glucose downstream and unmasks the transport capacity of SGLT1 (Fig. 8.1). This is also consistent with a high maximal glucose transport rate proposed for human SGLT1 based on in vitro studies.²⁸ As a consequence, combined inhibition of renal SGLT2 and SGLT1 is more glucosuric than inhibition of SGLT2 alone. This has been observed in nondiabetic and diabetic mice^{24,29} and in studies using a potent dual SGLT2/SGLT1 inhibitor.³⁰

The studies further indicated that SGLT2 and SGLT1 can account for all net glucose reabsorption in the kidney under normoglycemic conditions²⁴ (Fig. 8.3). Moreover, the data allowed estimating that the basal overall glucose reabsorption capacities for SGLT2 versus SGLT1 in a nondiabetic mouse kidney is in the range of 3:1 to 5:1.³¹

Clinical Relevance

Inhibition of Renal Glucose Reabsorption as a New Antihyperglycemic Therapy

SGLT2 inhibitors are a new class of antihyperglycemic drugs that have been approved in T2DM, work independently of insulin, improve glycemic control in all stages of diabetes mellitus in the absence of clinically relevant hypoglycemia, and can be combined with other antidiabetic agents. SGLT2 inhibition lowers glomerular hyperfiltration by a blood glucose-independent mechanism. By acting as a diuretic and lowering blood pressure and diabetic glomerular hyperfiltration, SGLT2 inhibitors have the potential to induce protective effects on the kidney and cardiovascular system beyond blood glucose control.

MOLECULAR CHARACTERIZATION OF PROXIMAL TUBULAR GLUCOSE TRANSPORT

APICAL GLUCOSE TRANSPORTERS

Crane first proposed in 1960 that active glucose transport in the intestinal epithelium (which expresses SGLT1) was energized by the Na^+ gradient across cell membranes, the so-called Na^+ -glucose cotransport hypothesis (for review, see Wright et al.²¹). The Na^+/K^+ -ATPase located on the basolateral membrane is the primary active and ATP-consuming transport step, which lowers cytosolic Na^+ concentrations and establishes the concentration gradient that drives Na^+ uptake, and, secondary, the uptake of other molecules from the luminal surface into proximal tubule cells (Figs. 8.1 and 8.4). This concept was rapidly refined and extended to active transport processes of a diverse range of molecules and ions including Na^+ -glucose cotransport in the kidney.²¹

SGLT1 and SGLT2 have been the most intensively studied members of the human SLC5 solute carrier (SLC) family, which now includes 12 members. Six of these are named as SGLTs, varying in their preferences for binding of glucose, galactose, mannose, fructose, myoinositol, choline, short-chain fatty acids, and other anions.²¹ All SGLTs have 15 exons, spanning from 8 to 72 kb, which code for 60- to 80-kDa proteins composed of 580–718 amino acids.²¹ The molecular nature of SGLTs has been largely pioneered by studies in

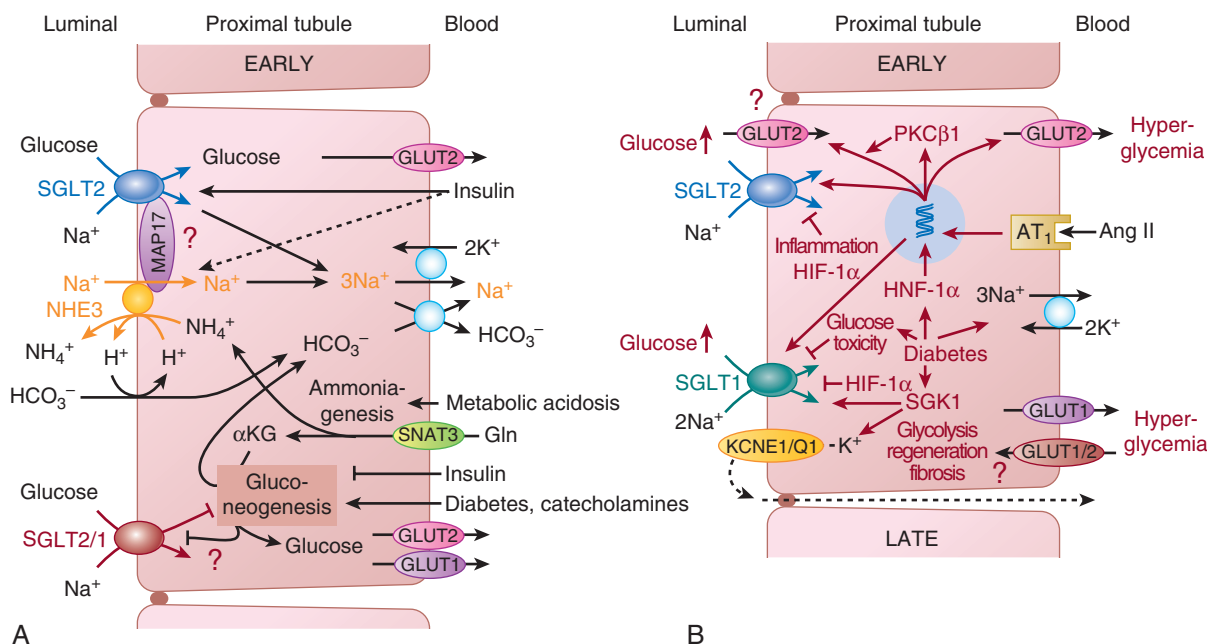


Fig. 8.4 Regulation of glucose transport in the proximal tubule. (A) Insulin is a physiologic stimulator of SGLT2, which may serve to maximize renal glucose reabsorption capacity in situations of increased blood glucose levels (e.g., after a meal). At the same time, enhanced Na^+ -glucose uptake and insulin suppress renal gluconeogenesis. The latter, in contrast, is stimulated by fasting, which may involve increased catecholamine levels. In metabolic acidosis, the increase in gluconeogenesis from glutamine (Gln) is linked to the formation of (1) ammonium (NH_4^+), a renally excreted acid equivalent, and (2) new bicarbonate, which is taken up into the circulation. The Na^+/H^+ exchanger NHE3 contributes to apical H^+/NH_4^+ secretion and $\text{Na}^+/\text{bicarbonate}$ reabsorption. SGLT2 and NHE3 are both stimulated by insulin to enhance Na^+ and glucose reabsorption, and their functions may be positively linked through the scaffolding protein MAP17. (B) Diabetes increases luminal glucose delivery to both SGLT2- and SGLT1-expressing segments. Glucose transporters GLUT2 and GLUT1 mediate glucose transport across the basolateral membrane, but GLUT2 may also translocate to the apical membrane in diabetes. Angiotensin II (Ang II), serum, glucocorticoid-inducible kinase SGK1, hepatocyte nuclear factor HNF-1 α , and protein kinase C PKCβ1 promote glucose reabsorption in the diabetic kidney, whereas hypoxia-induced HIF-1 α , inflammation, and excessive intracellular glucose levels may be inhibitory. Basolateral glucose uptake via GLUT1/2 may be involved in glycolysis and tubule regeneration after injury as well as hyperglycemia-induced TGF- β .

the laboratory of Wright and colleagues (see Wright et al.²¹ for review). This involved identifying and cloning SGLT1, identifying that defects in SGLT1 were associated with intestinal malabsorption of glucose–galactose and cloning of SGLT2. Wright's group also defined the crystal structure of a sodium galactose bacterial isoform in *Vibrio parahaemolyticus* (vSGLT), which allowed better characterization of how Na⁺ and sugar transport is coupled: Na⁺ binds first to the outside of the transport protein to open the outside gate, thereby permitting outside sugar to bind and be trapped; this is followed by a conformational change and the subsequent opening of the inward gate releases the Na⁺ and sugar into the cell cytoplasm. The transport cycle is completed by the change in conformation from an inward-facing ligand-free state to an outward-facing ligand-free state.^{21,32}

The sugar selectivity and transport kinetics of cloned SGLTs were determined using electrophysiologic techniques in various expression systems. The affinity of SGLT1 is similar for glucose and galactose, whereas SGLT2 does not transport galactose, and neither transports fructose.²¹ More recent studies in transfected human embryonic kidney (HEK) 293T cells indicated that the apparent affinities (K_m) for D-glucose are rather similar for human SGLT1 and human SGLT2, with values of 2 mM and 5 mM, respectively.²⁸ Sugar binding occurs in a Na⁺-dependent manner, and the K_m values for Na⁺ transport by human SGLT1 and human SGLT2 are 70 mM and 25 mM, respectively.²⁸ Thus, under euglycemic conditions, the glucose concentration in the tubular fluid of the very early proximal tubule (reflecting blood glucose levels) is similar to the K_m of SGLT2, whereas the luminal Na⁺ concentration of ~140 mM is higher than the K_m of SGLT2 and not rate limiting.

SGLT1 and SGLT2 transport Na⁺ and glucose with an Na⁺–glucose coupling ratio of 2:1 and 1:1, respectively.²⁸ The greater Na⁺–glucose coupling ratio of SGLT1 enhances its glucose concentration power and thereby the ability of the late proximal tubule to effectively reabsorb glucose despite falling luminal glucose concentrations (see Fig. 8.1). Na⁺–glucose transport is electrogenic, and the membrane potential and driving force are maintained by paracellular Cl[−] reabsorption or transcellular K⁺ secretion, the latter involving KCNE1/KCNQ1 channels in the luminal membrane of the proximal tubule^{33,34} (Fig. 8.1).

The mRNA expression of three other members of the SLC5 family that can transport glucose based on in vitro substrate studies has been detected in the kidney.³⁵ SGLT3 (*SLC5A4*) is not a glucose transporter, but glucose can depolarize the plasma membrane in its presence in a saturable, Na⁺-dependent manner, and this effect is inhibited by phlozizin. As such, it has been proposed to be a glucose sensor; however, its expression and function in the kidney remain unclear.³⁶ SGLT4 (*SLC5A9*) is expressed in the kidney, and transports glucose in COS-7 cells but with a lower apparent affinity than mannose (K_i 8 vs. 0.15 mM).³⁷ Thus, SGLT4 may primarily be involved in mannose homeostasis. SGLT5 (*SLC5A10*) is an Na⁺-dependent sugar transporter that has a relatively high affinity and capacity for mannose and fructose relative to glucose and galactose.^{38,39} SGLT5 mRNA is highly kidney abundant and expressed in kidney cortex,^{38,40} and recent studies in knockout mice indicated that SGLT5 is the major luminal transporter for fructose reabsorption in the kidney.⁴¹

BASOLATERAL GLUCOSE TRANSPORTERS

In the healthy kidney, the glucose that is being reabsorbed by proximal tubule cells is not linked to appreciable glucose metabolism in these cells. This is due to the fact that most glucose is reabsorbed in the early PCT (S1 segment), but these cells lack significant capacity for aerobic and anaerobic glycolysis.^{42–44} Thus, glucose that is taken up across the luminal membrane or formed within proximal tubule cells (see later) exits across the basolateral membrane into the interstitium by concentration-driven facilitative glucose transporters, GLUT2 and GLUT1 (see Fig. 8.1) and subsequently enters the peritubular capillaries by convection through fenestrated endothelial cells. GLUT2, the low-affinity (K_m ; 15–20 mM) “liver” transporter is primarily expressed in the PCT (S1/S2 segments), but GLUT2 mRNA has also been detected in the proximal straight tubule (PST) (S3 segment).⁴⁵ GLUT2 is thought to be the dominant transporter involved in basolateral exit of glucose derived from apical glucose uptake or gluconeogenesis in the PCT.^{46–48} In comparison, GLUT1, the high-affinity (1–2 mM) “erythroid/brain” transporter is expressed along the entire proximal tubule and has been implicated in transcellular glucose transport, particularly in the S3 segments.^{46–48} Notably, GLUT1 is also expressed in the basolateral membrane of further distal tubule segments and at higher levels than in S3 segments. This includes expression in the medullary thin and thick ascending limbs (TAL) of the rat kidney and at the highest levels in connecting segments and collecting ducts. In the latter, GLUT1 was expressed at the highest level in intercalated cells and to a lesser extent in principal cells.⁴⁸ These findings indicated a good correlation between the level of GLUT1 expression and the glycolytic activity of the different nephron segments, indicating that, in particular, the more distal tubule segments are taking up glucose for energy supply via basolateral GLUT1. Recent studies used positron emission tomography and α -methyl-4-[F-18]-fluoro-4-deoxy-d-glucopyranoside to monitor glucose transport in mouse kidneys lacking either SGLT1, SGLT2, or GLUT2. The studies confirmed prominent contributions of SGLT2 and SGLT1 to renal glucose uptake. Moreover, renal glucose reabsorption appeared absent in mice lacking GLUT2, consistent with a more prominent role of GLUT2 versus GLUT1 with regard to basolateral glucose exit of glucose in the proximal tubule⁴⁹ (Fig. 8.1). This is in line with the renal phenotype of patients with mutations in GLUT2 and GLUT1. Loss of function mutations in GLUT2 are the basis of the Fanconi–Bickel syndrome, which includes a renal Fanconi syndrome, a proximal tubulopathy consisting of glycosuria, phosphaturia, aminoaciduria, proteinuria, and hyperuricemia.^{50–52} The observed proximal tubulopathy may be due to intracellular glucose accumulation and glucotoxicity that occurs when basolateral glucose exit is blocked. In comparison, patients with GLUT1 mutations have primarily neurologic symptoms, and no renal phenotype has been documented.^{50,53}

In addition to GLUT1 and GLUT2, some of the other 12 members of the SLC2 gene family have been found in the kidney and may contribute to glucose transport, but little is known about their quantitative contribution.⁵⁴ For example, GLUT4 mRNA and immunoreactivity were focally localized in the TALH of the loop of Henle, coexpressed with IGF-I and increased by vasopressin treatment, indicating a potential

role in local fuel control.⁴⁵ GLUT5 is strongly expressed in the apical membrane of the S3 segment in the rat kidney but proposed to transport primarily fructose.^{45,55} GLUT12 can transport glucose, and an apical localization has been reported in distal tubules and collecting ducts, but its quantitative role remains to be determined.⁵⁶

RENAL FORMATION OF GLUCOSE

The kidneys not only reabsorb the filtered glucose and use glucose as an energy source, but they also generate new glucose. Gluconeogenesis involves the formation of glucose-6-phosphate from precursors such as lactate, glutamine, alanine, and glycerol with its subsequent hydrolysis by glucose-6-phosphatase to generate free glucose that can exit the cell. The healthy human kidneys produce ~15–55 g glucose per day by gluconeogenesis. In fact, the human liver and kidneys provide about equal amounts of glucose via gluconeogenesis in the postabsorptive state (i.e., 12–16 hours after the last meal).⁵⁷ Renal gluconeogenesis occurs along the entire proximal tubule, but its activity is usually higher in the earlier segments.^{43,58}

Renal gluconeogenesis is stimulated by epinephrine and inhibited by insulin⁵⁷ (see Fig. 8.4). Insulin-induced suppression of gluconeogenic gene expression in the proximal tubule was accompanied by phosphorylation and inactivation of forkhead box transcription factor 1 (FoxO1).⁵⁹ In contrast to the liver, renal gluconeogenesis is probably insensitive to glucagon.⁵⁷ Studies in humans indicated that in the postabsorptive state, renal gluconeogenesis uses primarily lactate as substrate, followed by glutamine, glycerol, and alanine.⁶⁰

In contrast to the uniform stimulation of gluconeogenesis along the entire proximal tubule by starvation, metabolic acidosis enhances gluconeogenesis primarily in S1 and S2 segments.^{43,58,61} Furthermore, gluconeogenesis in response to metabolic acidosis primarily uses glutamine as substrate. During the process of renal gluconeogenesis from glutamine, the conversion of glutamine to glutamate and α -ketoglutarate produces ammonium (NH_4^+), which is excreted into the urine as an acid equivalent. The subsequent pathway from α -ketoglutarate to glucose forms new bicarbonate, which is returned as buffer to the systemic circulation (see Fig. 8.4). The described link between proximal tubular ammonium, bicarbonate, and glucose formation explains why acidosis is a prominent stimulus for renal gluconeogenesis.^{57,60}

Apical glucose uptake via SGLT1 or SGLT2 can have an inhibitory influence on the expression of renal gluconeogenic genes (see Fig. 8.4). This effect may serve to prevent glucose overload in the cells of the proximal tubule, and has been proposed to involve glucose-induced and sirtuin 1–mediated deacetylation of peroxisome proliferator–activated receptor gamma coactivator 1- α , a coactivator of FoxO1.⁵⁹

In general, it is expected that cytosolic glucose in proximal tubule S3 segments is used for metabolism or leaves the cell via basolateral GLUT1. It has also been hypothesized that glucose generated from lactate in the medullary S3 segment forms part of an intrarenal Cori cycle⁶²; glucose enters the lumen by reversed transport through SGLT1 and is taken up by downstream tubular segments, where it is used as energy substrate for glycolysis (e.g., in medullary TAL), and the formed lactate is being returned to the neighboring S3 segment as a substrate for gluconeogenesis. Studies in human proximal tubule segments indicated that in contrast to S1

segments, lactate appears to be a better gluconeogenic precursor than glutamine in S2 and S3 segments.⁶³ Moreover, studies in mice indicated that the luminal membrane of the thick ascending limb in the outer stripe of the outer medulla and in the cortex (including the macula densa) express SGLT1.¹⁹ The rat appears to express SGLT1 also in the cortical TAL and macula densa.¹⁷ Further studies are needed to clarify whether the TAL (and macula densa) in human also express SGLT1, and determine the potential role of SGLT1 in these structures, including a proposed Cori cycle.

Taken together and under normal conditions, the PCT is the site of greatest renal glucose reabsorption and generation. The ability of early proximal tubular segments for gluconeogenesis but inability to metabolize glucose prevents a futile cycle. The reabsorption of filtered glucose and renal gluconeogenesis provide an energy source to distal tubular segments, primarily in the renal medulla, and returning the glucose to the systemic circulation helps to maintain blood glucose levels, particularly in the postabsorptive state. In addition, renal gluconeogenesis is closely linked to the renal response to metabolic acidosis. See Chapter 5 for additional discussion of the role of gluconeogenesis in renal metabolism.

RENAL GLUCOSE TRANSPORT IN DISEASE STATES

GLUCOSE TRANSPORT IS INCREASED IN THE DIABETIC KIDNEY

Lowering hyperglycemia is vital in diabetic patients to attenuate the progression of the underlying metabolic dysfunction⁶⁴ and to reduce the risk of diabetic complications including nephropathy and cardiovascular disease.⁶⁵ Current therapies for type 2 diabetes mellitus (T2DM) include drugs that target the liver, small intestine, adipose tissue, skeletal muscle, and/or pancreas. Many of these therapies, including insulin, have difficulties to establish adequate glycemic control without the potential for relevant unwanted side effects, including hypoglycemia and weight gain, and may not reduce cardiovascular complications.⁶⁶ The following sections outline how glucose transport changes in the diabetic kidney, its implications for diabetic kidney function, and how targeting renal glucose transport can serve as a new antihyperglycemic therapy.

Diabetes mellitus is associated with increased blood glucose levels. This enhances the amounts of glucose filtered by the kidneys, as long as GFR is preserved. In fact, the early phase of diabetes is often associated with an increase in GFR or glomerular hyperfiltration (see later), which further increases the tubular glucose load. At the same time, the tubular capacity to reabsorb glucose increases by ~20%–30% to ~500–600 g/day in patients with T2DM^{67,68} and type 1 diabetes (T1DM).³ Thus, diabetes often increases the glomerular filtration and tubular reabsorption of glucose. Moreover and despite increased blood glucose levels, diabetes also enhances renal gluconeogenesis.⁵⁷ The latter can be the consequence of diabetes-associated metabolic acidosis, and the induced gluconeogenesis involves metabolism of glutamine to glucose associated with the generation of ammonia and bicarbonate⁵⁷ (see Fig. 8.4). Other potential triggers for gluconeogenesis in diabetes include the activation of the sympathetic nervous system, the reduced insulin levels observed in T1DM, enhanced circulating fatty acids, or the kidneys receive another

signal from insulin-dependent cells that cannot take up glucose due to low insulin levels (T1DM) or insulin resistance (T2DM) and are glucose “starved.”

Increasing renal glucose reabsorption in response to a rise in filtered glucose makes sense with regard to energy substrate conservation. Moreover, the further distal segments may need/use more glucose as an energy substrate to reabsorb the load of salt and other compounds, which is increased due to glomerular hyperfiltration. Renal glucose retention and enhanced glucose formation become maladaptive in diabetes, however, when they sustain hyperglycemia (see Fig. 8.4). In this regard, the kidney provides a safety valve that can prevent extreme hyperglycemia. When blood glucose levels increase to the point that the filtered load exceeds the T_m or tubular transport capacity for glucose, then the surplus is excreted in the urine. The safety valve, however, only opens at rather high blood glucose levels (>15 mmol/L) (see Fig. 8.2) and only works as long as glomerular filtration is maintained, and its threshold depends on the level of expression and activity of the involved glucose transporter, and its functionality may thus vary from patient to patient.

GLUCOSE TRANSPORTERS IN THE DIABETIC KIDNEY

The levels of protein expression and activity of SGLT2, SGLT1, GLUT2, and potentially GLUT1 determine the capacity of renal glucose reabsorption, and their upregulation may explain the increased glucose transport maximum that can be observed in diabetes. The available preclinical and human studies reported increased, unchanged, or reduced renal glucose transporter expression and/or activity in diabetes or under high-glucose conditions.⁶⁹ The observed different responses may reflect different diabetes models, metabolic states, levels of kidney injury, other factors that regulate the expression of these transporters, the use of nonselective antibodies, or dissociations between mRNA and protein expression.

SGLT2 AND GLUT2

Using knockout mice as gold standard negative antibody controls, the renal protein expression of SGLT2 was found to be increased by 40%–80% in the early hyperglycemic stages of genetic mouse models of T2DM (db/db) and T1DM (Akita).^{70,71} The proximal tubule, like many nephron segments, expresses insulin receptors and binds insulin.⁷² Application of insulin to HEK-293T cells phosphorylated SGLT2 at Ser624, which increased Na^+ –glucose transport.⁷³ Thus, the insulin release following a meal may act on the PCT to enhance SGLT2 activity and conserve filtered glucose (see Fig. 8.4). Moreover, hyperinsulinemia associated with insulin resistance in obesity and T2DM may enhance renal SGLT2 activity⁷² (Fig. 8.4). This may be coordinated with a stimulatory effect of insulin on other Na^+ –coupled transporters in the proximal tubule, including the Na^+ –proton exchanger NHE3.^{72,74} Recent studies indicated that SGLT2 may be functionally linked to NHE3^{75–77} such that SGLT2 inhibition may, to some extent, inhibit NHE3 in the proximal tubule (Fig. 8.4). A similar interaction has been proposed for the coregulation of SGLT1 and NHE3 in the small intestine.

Consistent with a potential concerted regulation of luminal and basolateral glucose transport, upregulation of GLUT2 expression has been reported in renal proximal tubules in diabetic rats.^{78–81} Notably, studies in streptozotocin (STZ)-

induced T1DM in rats proposed targeting of GLUT2 (but not GLUT1) also to the brush border membrane of proximal tubules.^{82,83} The latter may be linked to protein kinase C $\text{PKC}\beta 1$ activation^{83–85} and may implicate facilitative glucose transport, together with SGLT2 and SGLT1 in the increased glucose reabsorption across the apical membrane of proximal tubules in the diabetic kidney (Fig. 8.4).

The available data on changes in glucose transporters in diabetic patients are sparse and also variable. Primary cultures of human exfoliated proximal tubular epithelial cells harvested from fresh urine of patients with T2DM showed an increased glucose uptake associated with increased protein expression of SGLT2 and GLUT2.⁸⁶ An increase in SGLT2 protein expression has also been reported in fresh kidney biopsies of patients with T2DM and advanced nephropathy.⁸⁷ On the other hand, the mRNA expression of SGLT2 and GLUT2 was slightly lower in 19 patients with T2DM and preserved kidney function as compared with 20 nondiabetic patients matched for age and estimated GFR (eGFR), all being subjected to nephrectomy.⁸⁸ Similar results were reported for SGLT2 and GLUT2 mRNA in another set of patients with T2DM, but the results did not reach statistical significance.⁸⁹

If an increase in SGLT2 expression occurs in the diabetic kidney, then it may simply reflect overall growth and hypertrophy of the diabetic proximal tubule and the associated increase in transport machinery,^{90,91} and this may be exaggerated with advanced nephropathy when nephrons are lost and the remaining nephrons aim to compensate. Moreover, upregulation of SGLT2 expression in diabetic rats has been linked to activation of Ang II AT1 receptors⁹² and the transcription factor, hepatocyte nuclear factor HNF-1 α .⁹³ The latter as well as HNF-3 β have also been implicated in renal GLUT2 upregulation⁷⁹ (see Fig. 8.4). Notably, pharmacologic inhibition of SGLT2 in normoglycemic mice also increased renal membrane SGLT2 protein expression,⁷¹ possibly reflecting a negative feedback regulation of SGLT2 expression by intracellular glucose levels. Along this line, if renal SGLT2 expression is reduced in the diabetic kidney, this may be due to enhanced diabetes-induced proximal tubular gluconeogenesis (Fig. 8.4) or reflect more severe tubular hypoxia or inflammation.^{94–96}

SGLT1 AND GLUT1

The renal expression of SGLT1 protein appears to vary among genetic mouse models of diabetes: renal SGLT1 protein expression was found to be increased in leptin-deficient ob/ob mice,⁹⁷ a model of T2DM, and reduced in Akita mice, a model of T1DM⁷⁰; the latter study used knockout mice as negative antibody control. In contrast to SGLT2 (see earlier), insulin stimulation slightly decreased SGLT1-mediated Na^+ –glucose transport in HEK-293T cells,⁷³ indicating differences in the regulation of these two transporters. In contrast to the strong increase in SGLT2 (see earlier), SGLT1 protein was not significantly changed in fresh kidney biopsies of patients with T2DM and nephropathy in comparison with nondiabetic controls.⁸⁷ The interpretation of renal SGLT1 mRNA expression data may be complicated by the observation that mRNA and protein expression can dissociate, at least in mouse kidney.⁹⁸

GLUT1 protein expression was downregulated in proximal tubules isolated from rat cortices at 2 and 4 weeks after STZ,⁸¹ but increased in kidneys of rats at 30 weeks after STZ.⁵⁶ A

study in patients with T2DM and preserved kidney function reported that in whole renal tissue, GLUT1 mRNA expression were slightly lower as compared with nondiabetic patients.⁸⁸

Why should diabetes reduce renal SGLT1 expression? Although this would make the renal glucose valve to open earlier (and make SGLT2 inhibitors more efficacious, see later), this may not be the kidneys' intention. A reduced renal SGLT1 protein expression was also observed in response to genetic or pharmacologic SGLT2 inhibition in nondiabetic mice.^{15,71} These conditions and diabetes share an enhanced glucose load to the late proximal tubule. In vitro studies in proximal tubule cells indicated that high glucose can reduce SGLT expression and Na⁺-glucose cotransport activity via enhanced oxidative stress.⁹⁹ Studies in a model of pig epithelial tubular cells (LLC-PK1) showed that hypoxia can diminish SGLT1 (and SGLT2) protein expression by activation of hypoxia-inducible factor-1 α (HIF-1 α).⁹⁵ Thus, an increased glucose load to the outer medullary S3 segment enhances Na⁺-glucose reabsorption and thereby hypoxia, which may downregulate SGLT1 to limit oxygen-consuming transport work and glucotoxicity in this segment, which has a high sensitivity to acute injury⁹⁴ (see Fig. 8.4).

In comparison, an increase in SGLT1 expression in the diabetic kidney would further increase the renal glucose reabsorption capacity but may put the S3 segment at risk of hypoxia and enhanced glucotoxicity. Studies in Akita diabetic mice indicated that the serum and glucocorticoid-inducible kinase SGK1 may stimulate SGLT1 activity and glucose reabsorption in PSTs.¹⁰⁰ SGK1 could also promote proximal tubular glucose reabsorption by enhancing the activity of luminal K⁺ channels (KCNE1/KCNQ1), which maintain the electrical driving force during electrogenic Na⁺-glucose cotransport^{33,34,101} (see Fig. 8.4). SGK1 was upregulated in proximal tubules in patients with diabetic nephropathy.¹⁰²

Transport functions in proximal tubules require high turnover of ATP, which, under normal conditions, is derived primarily through mitochondrial oxidative phosphorylation.^{43,103} This may change in pathophysiologic situations with impaired mitochondrial function, when glycolysis may be enhanced and contribute to maintaining ATP. For example, a shift to glycolysis has been proposed in proximal tubules regenerating from acute kidney injury (AKI) as well as proximal tubules undergoing atrophy.¹⁰⁴ This metabolic switch to glycolysis occurred early during proximal tubule regeneration and was reversed during successful tubular recovery, but persisted and became progressively more severe in tubule cells that failed to redifferentiate. Tubular upregulation of HIF-1 α in mice enhanced renal GLUT1 mRNA expression; this was associated with less oxygen consumption and increased glycolysis.¹⁰⁵ Thus, hypoxia may increase basolateral GLUT1-mediated facilitative uptake of glucose, which is then used for glycolysis and recovery. Hypoxia-induced GLUT1 likely applies to distal tubule segments but may also be relevant for medullary S3 segments.¹⁰⁴ In this regard, studies in the proximal tubular cell line LLC-PK1, which was cultured and polarized on porous tissue culture inserts, showed that basolateral exposure to 25 mmol/L D-glucose enhanced glucose uptake via GLUT1 and the subsequent intracellular metabolism of glucose enhanced TGF- β 1 synthesis and secretion; this was not observed in response to apical glucose exposure.¹⁰⁶ These in vitro studies suggest that it may be the hyperglycemia-induced persistent uptake of glucose via

basolateral GLUT1 (or GLUT2), rather than the filtered glucose, that affects the tubular synthesis of TGF- β 1 and thereby the development of tubulointerstitial fibrosis and tubular growth (Fig. 8.4).

INHIBITION OF RENAL GLUCOSE REABSORPTION AS A NEW ANTIHYPERGLYCEMIC THERAPY

When blood glucose levels increase to the point that the filtered load exceeds the transport capacity of the tubular system, then the surplus is excreted in the urine. This renal safety valve can prevent extreme hyperglycemia. As outlined earlier, most capacity for renal glucose reabsorption is provided by SGLT2 in the early proximal tubule. When SGLT2 is inhibited, the reabsorptive capacity for glucose declines to the residual capacity of SGLT1, which equals ~80 g/day. In other words, SGLT2 inhibition causes the renal safety valve to open at a lower threshold (see Fig. 8.2) and makes it also relevant to glucose homeostasis in the euglycemic and moderately hyperglycemic range. Several SGLT2 inhibitors have now been approved as glucose-lowering agents for subjects with T2DM and preserved kidney function. Previously, the presence of glucosuria in a diabetic patient indicated inappropriate blood glucose control, as it showed that blood glucose was so high that the filtered glucose overwhelmed the glucose reabsorption capacity. In contrast and with the use of SGLT2 inhibitors, glucosuria is purposely induced to improve blood glucose control. The following sections discuss the role of SGLT2 in the pathophysiology of renal glucose reabsorption and outline the unexpected logic of inhibiting SGLT2 in the diabetic kidney.⁹⁰ This includes the counterproductive enhancement of renal glucose reabsorption via SGLT2 in diabetes as well as a brief discussion of the basic mechanisms that link a primary inhibition of Na⁺-glucose cotransport in the kidney to secondary beneficial consequences on the metabolism, the kidneys, and the cardiovascular system.

Long-term access to excessive exogenous energy resources is not part of human evolution. As a consequence, it may not come as a surprise that the body's responses to excess exogenous energy resources can be maladaptive. In contrast, the body's responses to environments with scarce energy resources have been intensively tested and refined during evolution for the survival of the organism. Therefore, targeting metabolism in the "periphery" by inhibiting renal glucose reabsorption and spilling glucose as an energy resource into the urine and then using the central metabolic counterregulatory mechanisms to readjust the metabolism, may provide unique benefits as an antihyperglycemic approach.⁹⁰ This is supported by clinical outcome studies using an SGLT2 inhibitor on top of standard of care in patients with high cardiovascular risk that demonstrated protective effects with regard to clinically relevant renal and cardiovascular outcomes.^{107–109}

Phlorizin is a flavonoid contained in the bark of various fruit trees and was discovered to cause glucosuria more than 100 years ago.¹¹⁰ Phlorizin competitively inhibits SGLT2 and SGLT1, the former with a 10-fold higher affinity.^{21,28} SGLT1 is expressed in many other organs and is the primary pathway for glucose reabsorption in the intestine.¹¹¹ As a consequence, oral administration of phlorizin is burdened by extrarenal side effects, most prominently diarrhea. In comparison and in healthy subjects, SGLT2 appears to be expressed only in

kidney proximal tubule,^{18,98} with a proposed expression and function in α cells of the pancreas¹¹² needing confirmation. Inhibition of renal glucose transport became practical when phlorizin derivatives were developed that are specific for SGLT2, have good oral bioavailability, and are suitable for once-daily dosing.¹¹³ Three members in this class, dapagliflozin (Forxiga or Farxiga in the United States), canagliflozin (Invokana), and empagliflozin (Jardiance) are approved in the United States and Europe for use in T2DM with preserved kidney function. Others, including ipragliflozin (Suglat), luseogliflozin (Lusefi), and tofogliflozin (0) are approved in Japan. SGLT2 inhibitors are under clinical investigation as add-on therapies to insulin in T1DM.

SGLT2 inhibitors act on their target from the extracellular surface of the cell membrane¹¹⁴ and reach their target by glomerular filtration and, as indicated for empagliflozin, also by tubular secretion.¹¹⁵ SGLT2 inhibitors induce a sustained urinary glucose loss of 40–80 g/day.^{113,116,117} In patients with T2DM, this is associated with a decrease in Hb A1C levels by 0.5%–0.7% at 12 weeks of treatment, and this effect persisted for up to 52 weeks.¹¹⁸ The higher the blood glucose level and GFR, the more glucose is filtered and reabsorbed and, as a consequence, can be excreted in response to SGLT2 blockade. Thus, SGLT2 blockers naturally have a greater efficacy when it is desirable for them to be more efficacious.^{15,119,120} By lowering blood glucose levels and body weight, SGLT2 inhibitors improve β -cell function and sensitivity to insulin in patients and rodent models with T2DM.^{119,121–124} Because the renal mechanism of action of SGLT2 inhibitors is independent of insulin, their efficacy is not declining with progressive β -cell dysfunction and/or insulin resistance, and SGLT2 inhibitors act synergistically with other blood glucose-lowering agents.¹¹³

Two SGLT2 inhibitors have now been evaluated in major clinical trials in patients with T2DM: empagliflozin in the 7,020-patient EMPA-REG OUTCOME trial^{107,109} and, more recently, canagliflozin in 10,142 patients in the CANVAS program.¹⁰⁸ In addition to cardiovascular endpoints, both trials also included measurement of albuminuria and eGFR. The outcomes of the EMPA-REG OUTCOME and the CANVAS program are similar in most regards. Both went beyond the requisite safety parameters to show ~35% reductions in the incidence of heart failure. Both trials also reported beneficial effects on the kidney, including 40%–50% reductions in the hazard ratios for albuminuria or decline in eGFR. The relative risk of cardiovascular death was significantly reduced by SGLT2 inhibition in the EMPA-REG OUTCOME trial but not in the CANVAS program. This difference might be due to the higher prevalence of cardiovascular disease in the EMPA-REG OUTCOME cohort at baseline. The main cardiovascular effect of SGLT2 inhibition was on heart failure, rather than ischemic events, and both trials showed tangible benefits on heart failure outcomes. These benefits occurred when added to standard care, which included ~80% of patients being treated with an angiotensin-converting enzyme inhibitor or angiotensin AT1 receptor antagonist. The main side effect of SGLT2 inhibitors is an increased risk of genitourinary infections due to the glucosuric effect.¹²⁵

How can inhibition of renal glucose transport protect the kidney and cardiovascular system? By reducing hyperglycemia, SGLT2 inhibitors have the potential to reduce glucotoxicity in the kidney and extrarenal organs.^{126,127} In accordance,

studies in diabetic rodents have shown that SGLT2 inhibition can reduce growth, lipid accumulation, inflammation, and injury of the diabetic kidney secondary to a strong blood glucose-lowering effect^{70,71,87,97,128–131} (Fig. 8.5). The observed small effect of SGLT2 inhibitors on blood glucose control in the EMPA-REG OUTCOME trial and the CANVAS program alone, however, appears insufficient to fully explain the rapid beneficial effect on heart failure detectable within a few months. Although other mechanisms are likely to contribute (see later), it is also possible that, in contrast to SGLT2 inhibitors, these other agents have simultaneous countervailing effects that offset the benefits of better glycemic control, including increased obesity or an increased hypoglycemia risk.

SGLT2 inhibition lowers body weight and has a low hypoglycemia risk. In patients with T2DM, including those in the EMPA-REG OUTCOME trial and the CANVAS program, the glucosuric effect of SGLT2 inhibition was associated with a 2- to 3-kg lower body weight. Although the diuretic effect and fluid loss may contribute to the initial weight loss, the majority of the steady-state weight loss with SGLT2 inhibitor treatment is due to fat loss, including lesser visceral and subcutaneous fat^{132–134} (see Fig. 8.5), due to a shift in substrate utilization from carbohydrates to lipids.^{119,135,136} The released free fatty acids are used by the liver to form ketone bodies and thus increase ketogenesis.¹³⁷ SGLT2 inhibitors may improve cardiac outcomes in part by increasing plasma levels of ketone bodies like β -hydroxybutyrate, which are used as energy substrate to improve the performance of cardiac myocytes (or the kidney) in diabetes mellitus^{138,139} (Fig. 8.5). SGLT2 inhibitors can increase the risk of diabetic ketoacidosis,¹³⁷ particularly when the drugs are used off-label in patients with T1DM.¹³⁷

SGLT2 inhibitors do not increase the incidence of hypoglycemia.^{107–109,118} This is because they become ineffective at lowering blood glucose any further once the filtered glucose load falls to ~80 g/day, which can be handled by renal SGLT1 (see Fig. 8.1). In addition, SGLT2 inhibitors leave the metabolic counterregulation intact and increase plasma glucagon concentrations and subsequently endogenous hepatic glucose production (gluconeogenesis) in patients with T2DM.^{119,122} This is potentially relevant for cardiovascular outcome, because episodes of hypoglycemia can impair the cardioprotective effects of antihyperglycemic therapy.¹⁴⁰

SGLT2 inhibition lowers blood pressure and improves hyperuricemia. A meta-analysis of patients with T2DM treated with SGLT2 inhibitors found a consistent decrease in systolic blood pressure of 3–6 mm Hg,¹¹⁸ similar to preclinical data and the EMPA-REG OUTCOME trial and the CANVAS program. The magnitude of this blood pressure effect is expected to have cardiovascular protective consequences, particularly in high-risk patients.¹⁴¹ The blood pressure-lowering effect of SGLT2 inhibition relates to the reduction in body weight and a modest glucose-based osmotic diuresis (100–470 mL/day) and a small natriuretic effect.^{126,142–145} The lower blood pressure and an associated modest reduction in plasma volume¹⁴⁶ may quickly reduce cardiac pre- and afterload and thereby contribute to the rapid beneficial effects in heart failure patients¹⁰⁷ (see Fig. 8.5).

Beneficial renal and cardiovascular effects of SGLT2 inhibition may also be due to a plasma uric acid-lowering effect.¹⁴⁷ The uricosuric effect of SGLT2 inhibitors is positively

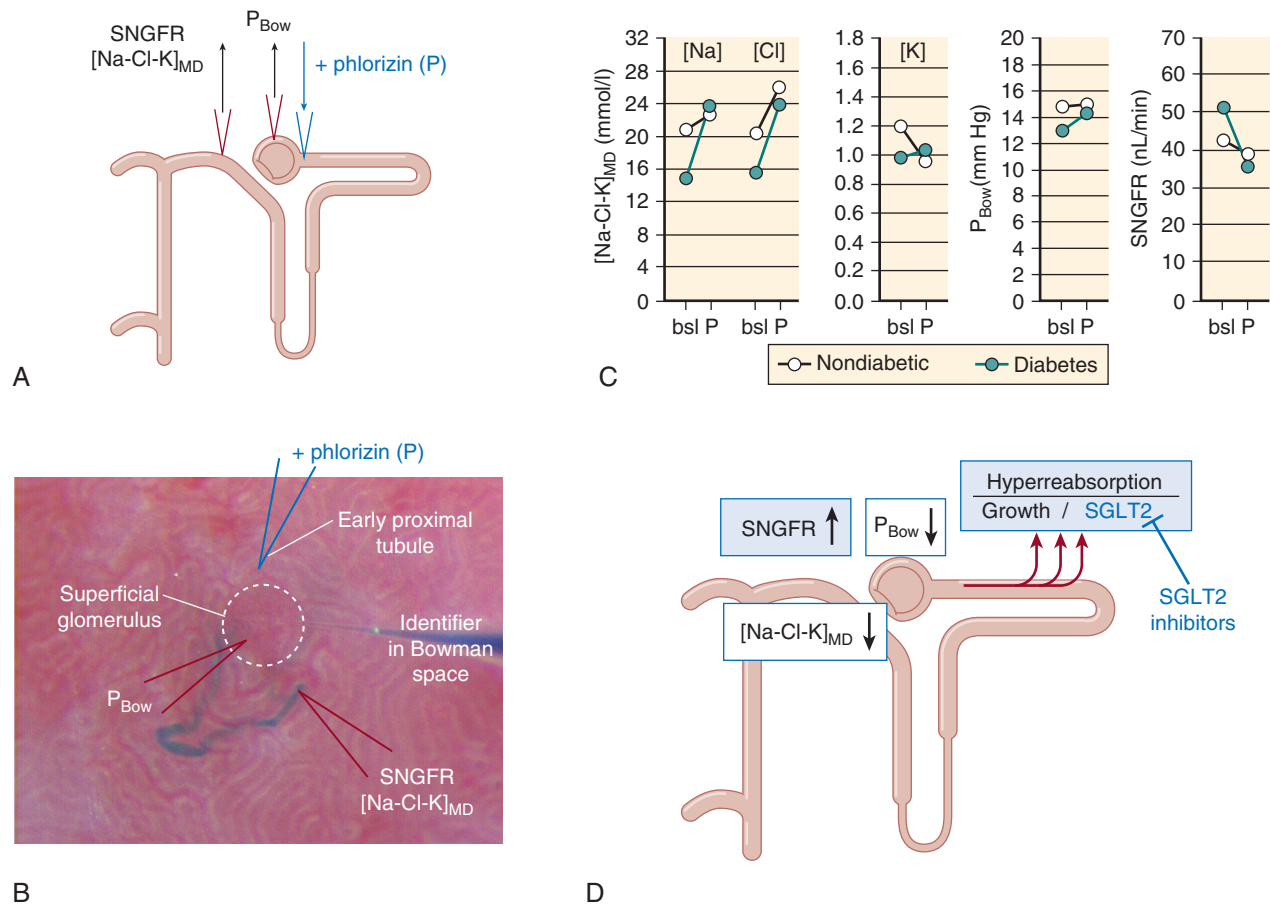


Fig. 8.6 The tubular hypothesis of diabetic glomerular hyperfiltration: effect of SGLT2 inhibition. (A, B) In vivo micropuncture studies in rats with superficial glomeruli were performed in nondiabetic and streptozotocin diabetic rats.¹⁵² Small amounts of blue dye were injected into the Bowman space to determine nephron configuration, including the first proximal tubular loop and the early distal tubule close to the macula densa. Tubular fluid was collected close to the macula densa to determine (1) the tubuloglomerular feedback signal ($[Na-Cl-K]_{MD}$), and (2) single-nephron glomerular filtration rate (SNGFR) by inulin clearance. The Bowman space was punctured to determine the hydrostatic pressure (P_{Bow}). Measurements were performed under control conditions and following application of the SGLT2/SGLT1 inhibitor phlorizin into the early proximal tubule (i.e., without changing systemic blood glucose levels). (C) Basal measurements (bsl) revealed that glomerular hyperfiltration in diabetes was associated with reductions in $[Na-Cl-K]_{MD}$ and P_{Bow} . Adding phlorizin (P) had a small effect in nondiabetic rats but normalized $[Na-Cl-K]_{MD}$, P_{Bow} , and SNGFR in diabetes. (D) Diabetes induces a primary hyperreabsorption in the proximal tubule due to tubular growth and enhanced Na^+ -glucose cotransport, which, through tubuloglomerular feedback ($[Na-Cl-K]_{MD}$) and reducing tubular back pressure (P_{Bow}), causes glomerular hyperfiltration. SGLT2 contributes to the tubular hyperreabsorption, and as a consequence, SGLT2 inhibition mitigates these changes and lowers glomerular hyperfiltration. (This figure was modified with permission from Vallon V, Thomson SC. Targeting renal glucose reabsorption to treat hyperglycaemia: the pleiotropic effects of SGLT2 inhibition. *Diabetologia* 2017;60:215–225.)

changes in tubular back pressure contribute equally to the increase in GFR in diabetes.¹⁵¹

Vice versa, SGLT2 inhibition attenuates proximal tubule hyperreabsorption in the diabetic kidney and thereby lowers diabetic glomerular hyperfiltration (see Figs. 8.5 and 8.6). This has been shown in micropuncture studies in rats using direct application of phlorizin into the Bowman space¹⁵² and by acute or chronic systemic application of selective SGLT2 inhibitors.¹⁵³ In accordance, pharmacologic or genetic inhibition of SGLT2 suppressed hyperfiltration on the whole-kidney level in diabetic mice.^{70,71} Consistent with the proposed local mechanism, the suppression of diabetic hyperfiltration in response to SGLT2 inhibition was associated with an increase in the NaCl concentration at the macula densa^{152,153} and in the hydrostatic pressure in the Bowman space,¹⁵² and was independent of effects on blood glucose^{70,152,153} (Fig. 8.5). The GFR-lowering effect has more recently been confirmed

in humans. The SGLT2 inhibitor empagliflozin decreased GFR by 19% in T1DM patients with baseline hyperfiltration independently of lowering blood glucose levels.¹²⁰ The SGLT2 inhibitor canagliflozin initially lowered eGFR in patients with T2DM and basal eGFR of ≥ 55 mL/min/1.73 m². Following this initial dip, eGFR increased over the following weeks and months in the canagliflozin-treated group such that eGFR was better preserved after 2 years of follow-up and associated with reduced urinary albumin-to-creatinine ratios than in the control group, which had been treated with glimepiride to achieve similar blood glucose control.¹⁵⁴

Surviving nephrons in advanced stages of CKD are assumed to hyperfilter as a way of compensation for the reduced nephron number and thus maintain a high glucose load on the level of the single nephron. This should preserve the acute GFR-lowering effect of SGLT2 inhibition, even if the effect on overall glucose homeostasis was attenuated. In

accordance, the SGLT2 inhibitor canagliflozin modestly reduced eGFR together with proteinuria within 3 weeks in patients with T2DM and basal eGFR values between 30 and 50 mL/min/1.73 m² (CKD3).¹⁵⁵ Empagliflozin also induced a small decline in eGFR in patients with T2DM and CKD2 and CKD3; this effect was maintained at 52 weeks, associated with reduced urine albumin-to-creatinine ratios and, most importantly, full reversibility after a 3-week washout period, indicating a functional GFR reduction.¹⁵⁶

Lowering single-nephron glomerular hyperfiltration in CKD and thereby the oxygen-consuming transport work may help to preserve the integrity of the remaining nephrons and overall kidney function in the long term (see Fig. 8.5). This has been proposed for blockers of angiotensin II¹⁵⁷ and may also apply to SGLT2 inhibitors. Because ~80% of patients were also treated with a form of angiotensin II blockade, the EMPA-REG OUTCOME trial and CANVAS program provided evidence that the two strategies are additive and apply to patients with initial GFRs of at least 30 mL/min/1.73 m² of body surface area.^{108,109} The additive effect is consistent with the concept that angiotensin II blockade is primarily dilating the efferent arteriole, whereas SGLT2 inhibition primarily constricts the afferent arteriole.

SGLT2 inhibition has distinct effects on renal cortical and medullary O₂ requirements. Mathematical modeling predicted that inhibition of SGLT2 in the diabetic kidney reduces oxygen consumption in the PCT and renal cortex, in part by lowering GFR^{142,143} (Fig. 8.5). The predicted increase in cortical O₂ pressure and availability has been observed in a diabetic rat model using phlorizin, a dual SGLT1/SGLT2 inhibitor.¹⁵⁸ Interestingly, preserving renal cortical oxygenation may be important to preserve kidney function in patients with CKD.¹⁵⁹

SGLT2 inhibition also shifts glucose uptake downstream to the S3 segments (see earlier) and enhances transcellular Na⁺ reabsorption in distal segments, including the S3 segment and medullary TAL. This may further reduce the already physiologically low O₂ availability in the renal outer medulla. The latter has been proposed for SGLT2 inhibition using mathematical modeling^{142,143} and was shown in vivo in rats in response to acute dual SGLT2/SGLT1 inhibition by phlorizin in nondiabetic and diabetic rats.¹⁵⁸ The effect on medullary transport and oxygenation would be attenuated by the reduction in blood glucose and GFR in response to SGLT2 inhibition.^{142,143} Moreover, the proposed SGLT2 inhibitor-induced reduction in oxygen pressure in the deep cortex and outer medulla may stimulate hypoxia-inducible factors HIF-1 and HIF-2 (see Fig. 8.5). Gene knockout of SGLT2 increased the renal mRNA expression of hemoxygenase,^{1,7} a tissue-protective gene that is induced by HIF-1 α . On the other hand, activation of HIF-2 may explain an enhanced erythropoietin release from renal interstitial cells in response to SGLT2 inhibition.¹⁶⁰ Together with the diuretic effect, the latter may contribute to the observed modest increase in hematocrit and hemoglobin in response to SGLT2 inhibition. This may improve the oxygenation of the kidney outer medulla and cortex but also facilitate oxygen delivery to the heart and other organs (Fig. 8.5). Notably, changes in hematocrit and hemoglobin from baseline explained 51.8% and 48.9%, respectively, of the effect of the SGLT2 inhibitor empagliflozin versus placebo on the risk of cardiovascular death.¹⁶¹ In other words and in addition to its volume effect, SGLT2 inhibition may simulate systemic hypoxia to the oxygen

sensor in the deep cortex and outer medulla of the kidney, and the induced response then helps the failing heart and also the kidney. In accordance with an overall nephroprotective effect, SGLT2 inhibitor use reduced the risk of AKI in an analysis of >3000 patients with T2DM by ~50%.¹⁶² Nevertheless, caution is warranted, as excessive volume depletion and the transport shift to the outer medulla may increase the AKI risk in individual sensitive patients.

Preservation of blood pressure-lowering and heart failure-protective effects of SGLT2 inhibitors in CKD despite attenuated antihyperglycemic effects. The amount of filtered glucose determines the glucosuric and blood glucose-lowering effect of SGLT2 inhibition. As a consequence, the antihyperglycemic effects of SGLT2 inhibitors are attenuated in patients with reduced GFR. In contrast, the blood pressure-lowering and heart failure-protective effects are preserved in patients with CKD and reduced GFR (eGFR $\geq$ 30 mL/min 1.73 m²).^{163,164} Modeling studies of CKD and nephron loss predicted that the increase in single-nephron GFR in remaining nephrons and the reduction of glucose reabsorption by SGLT2 inhibition increase paracellular Na⁺ secretion in the proximal tubule.¹⁶⁵ Thus, the model predicted that the chronic natriuretic and diuretic effects of SGLT2 inhibition persist in CKD. The modeling approach also predicted that the SGLT2 inhibition-induced changes in the oxygen signal at the renal sensor are preserved in CKD.¹⁶⁵

Ongoing trials with different SGLT2 inhibitors, also including studies in nondiabetic patients with heart failure and/or CKD, will provide further data for comparison between SGLT2 inhibitors and are expected to further refine our understanding of the therapeutic potential and safety of SGLT2 inhibition.

ORGANIC CATIONS AND ANIONS

ORGANIC ANION TRANSPORTERS AND ORGANIC CATION TRANSPORTERS

Historically, the renal organic anion transport system has been one of the best studied in physiology.^{166–173} Essentially, it has been operationally characterized as the probenecid-sensitive para-aminohippurate (PAH) transport system. Classically, it is described as a proximal tubule transport system of small organic anions (e.g., PAH) bound with low affinity to plasma proteins (mainly albumin). Because, with an intact glomerular filtration barrier, albumin-bound molecules would not be filtered, they move into the peritubular capillaries. Molecules like PAH are efficiently extracted on a “first pass” by a high-capacity transport system with selectivity for organic anions. This explains why PAH clearance can be used as a measure of renal plasma flow. Operationally, the system can be blocked by the organic anion drug probenecid, which has seen considerable clinical use in the setting of hyperuricemia and to increase blood levels of other organic anion drugs like penicillins and cidofovir.^{174–176}

The main gene responsible for this probenecid-inhibited PAH transport is a SLC transporter known as OAT1 (SLC22A6) and was originally called NKT (novel kidney transporter).^{177,178} Like most other members of the mammalian SLC family, it has 12 membrane-spanning segments (Fig. 8.7A). At the time of its discovery, NKT was proposed to function as either

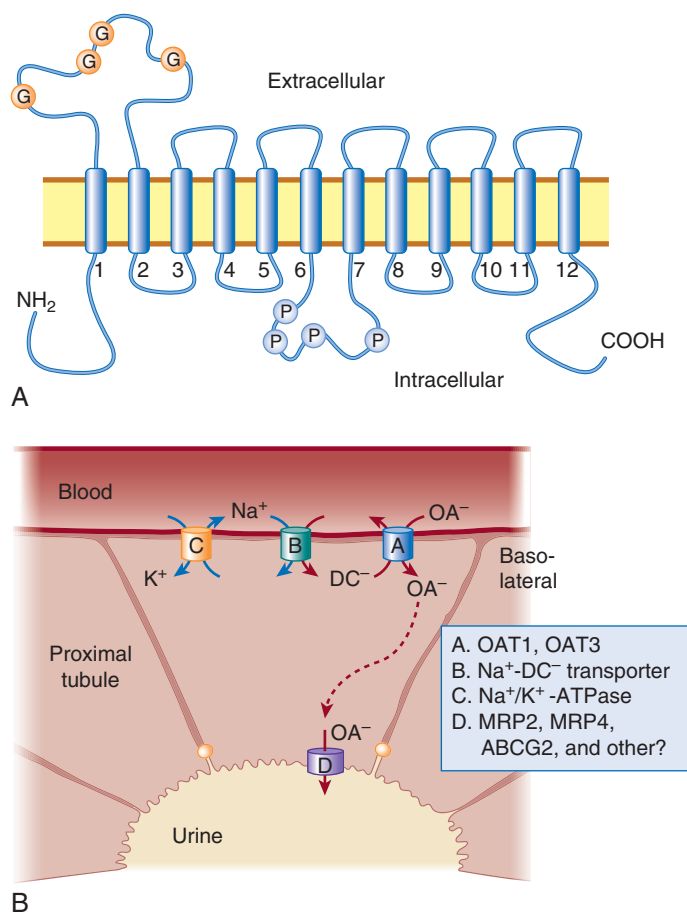


Fig. 8.7 Topology of OAT1 and schematic of OAT1-mediated organic anion influx uptake as a “tertiary” transport process.

(A) An intracellular loop links two six-transmembrane domains yielding 12 membrane-spanning regions similar to many other SLC transporters (G, glycosylation sites; P, PKC, phosphorylation sites). (B) Schematic depicting a proximal tubule cell showing OAT-mediated influx of organic anions (OA^-) from the plasma to the lumen. OAT1 (A), OA^- influx by OAT1 at the basolateral membrane via antiport of dicarboxylates (DC^-) down a gradient. OAT-mediated influx is connected to the transmembrane gradient of dicarboxylates created as a result Na^+ /dicarboxylate cotransporter and mitochondrial TCA cycle (B). This “tertiary” process depends upon the ATP hydrolyzed by the Na^+ - K^+ -ATPase to create an extracellular-to-intracellular sodium gradient (C). Transport of OA^- into urinary space (D) probably occurs through a number of apical membrane transporters, including the MRPs. (Modified from Nigam SK, Bush KT, Martovetsky G, et al. The organic anion transporter (OAT) family: a systems biology perspective. *Physiol Rev.* 2015;95:83–123; and Eraly SA, Bush KT, Sampogna RV, et al. The molecular pharmacology of organic anion transporters: from DNA to FDA? *Mol. Pharmacol.* 2004;65:479–487.)

an organic anion or organic cation transporter; numerous studies have since confirmed that, although OAT1 and other members of the OAT subfamily are predominantly organic anion transporters, they can also transport organic cations and zwitterions.^{179,180}

The current view of how a prototypical organic anion—PAH in this case—is taken up across the basolateral membrane (blood side) of the proximal tubule cell involves three different transporters: OAT1, the sodium–dicarboxylate cotransporter (NaDC3, SLC13A3), and the sodium–potassium ATPase (Fig. 8.7B). This “tertiary” transport system is believed to operate

through the following mechanisms: (1) exchange (antiport) of PAH in the plasma with α -ketoglutarate by OAT1; (2) cotransport (symport) of sodium and α -ketoglutarate into the cell by NaDC3; and (3) extrusion of sodium into the plasma, creating a sodium gradient, through the ATP-dependent action of the sodium–potassium ATPase (Na-K-ATPase). Thus, organic anion transport is ATP dependent only in an indirect way and depends on two other gradients: (1) a sodium gradient generated by the Na-K-ATPase, among other factors, that enables cotransport of sodium with α -ketoglutarate into the cell by NaDC3; and (2) a high intracellular α -ketoglutarate level that is partly due to the aforementioned NaDC3 cotransport and partly contributed by aerobic mitochondrial metabolism resulting in the generation of tricarboxylic acid (TCA) cycle intermediates such as α -ketoglutarate. Blocking any of these processes, for example, ouabain inhibition of the Na-K-ATPase, lithium inhibition of NaDC3, or probenecid inhibition of OAT1, markedly diminishes or even completely abolishes PAH transport via OAT1. In general, OAT3 appears to function via an analogous tertiary transport system, although it is not entirely clear that the linkage to α -ketoglutarate exchange is as tight.

When OAT1 was originally discovered (as NKT), two other homologous transporters, NLT (now OAT2, SLC22A7)¹⁸¹ and OCT1, organic cation transporter 1 (SLC22A1)¹⁸² were also found in the sequence database, and it was proposed that this was a new family of SLC transporters, which is now known as SLC22.¹⁷⁸ Since then, the SLC22 family has grown to roughly 30 transporters in humans and mice.^{183,184}

A new evolutionary-based classification of SLC22 has been proposed, which divides the family into a major OAT clade and a major OCT clade, which further divide into six subclades (Fig. 8.8). The assumption is that the various SLC22 subclades defined by phylogenetic relationships will lead to a better functional classification and help deorphanize SLC22 transporters of unclear function. Accordingly, subclades within the OAT clade include the OAT subclade, the OAT-like subclade, and the OAT-related subclade; within the OCT clade is the OCT subclade, OCT-like (OCTN) subclade, and the OCT-related subclade.^{183,185} The specific details of many of these transporters are beyond the scope of this chapter, and, indeed, many remain “orphans” with respect to endogenous substrate preference. Nevertheless, it is important to point out that, with the exception of certain subclades, the predominant expression for most of the other SLC22 transporters tends to be in the kidney proximal tubule, choroid plexus, or the liver. Even so, SLC22 transporters are expressed throughout the body, and there are even members with highly selective localization to the olfactory epithelium and brain substructures (OAT6, SLC22A20).¹⁸⁶ SLC22 is thus a very interesting SLC family involved in the transport of anionic, cationic, and zwitterionic drugs; toxins; metabolites; signaling molecules; antioxidants; dietary components; vitamins; gut microbiome products; and uremic toxins.¹⁸³

Here we focus on OAT1 (SLC22A6) and OAT3 (SLC22A8), which are the major renal metabolite, drug, and toxin (including uremic toxin) organic anion transporters (Table 8.1); OCT2 (SLC22A2), the major transporter of cationic drugs and metabolites; and URAT1 (SLC22A12), the latter being the most extensively studied of several urate transporters in the OAT subclade. Along with a number of other SLC and ABC transporters, these SLC22 transporters are the most

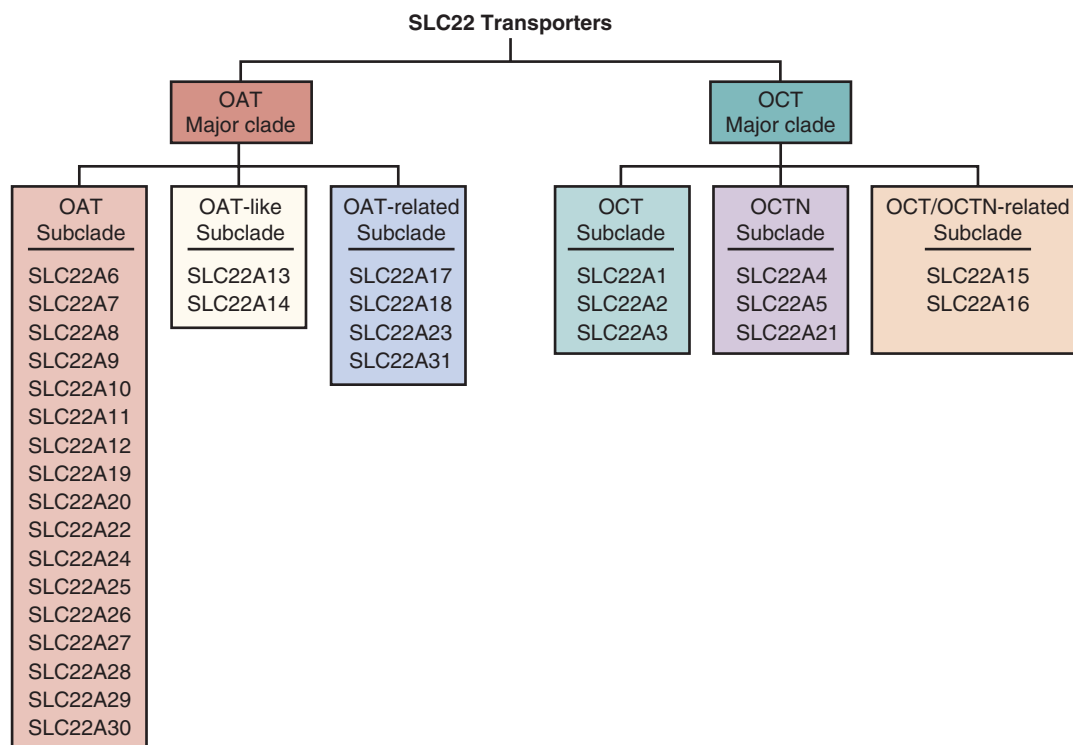


Fig. 8.8 The six subfamilies of SLC22 transporters. Evolutionary analysis indicates that SLC22 transporters are highly conserved and found in fly, worm, sea urchin, and other organisms. The SLC22 family is composed of two major clades, which are the organic anion transporter (OAT) major clade and organic cation transporter (OCT) major clade. Each of these clades is divided into three subclades, designated as OAT, OAT-like, OAT-related, OCT, OCTN (organic cation/carnitine transporter), and OCT/OCTN-related. (Modified from Nigam SK. The SLC22 transporter family: a paradigm for the impact of drug transporters on metabolic pathways, signaling, and disease. *Ann Rev Pharmacol Toxicol*. 2018;58:32.31–32.25; and Zhu C, Nigam KB, Date RC, et al. Evolutionary analysis and classification of OATS, OCTS, OCTNS, and other SLC22 transporters: structure-function implications and analysis of sequence motifs. *PLoS ONE*. 2015;10:e0140569.)

clinically relevant and quantitatively important organic anion and organic cation transporters in the proximal tubule of the kidney.^{184,187}

OAT1 (SLC22A6) AND OAT3 (SLC22A8)

Both OAT1 and OAT3 were among the original seven drug transporters that regulatory agencies identified as important for analysis of the possibility of transport of new drug entities.¹⁸⁸ This regulatory attention has perpetuated the notion that these transporters primarily transport drugs. Although it is true that the OATs transport drugs (e.g., antibiotics, antivirals, nonsteroidal antiinflammatory drugs, diuretics) and toxins (e.g., organic mercurials, aristolochic acid), it is now clear that they, as well as the other five SLC and ABC transporters highlighted by regulatory agencies, transport many endogenous metabolites, signaling molecules, vitamins, gut microbiome, and dietary products.¹⁸⁴

Indeed, there is a growing appreciation that the primary function of these multispecific transporters may not be the handling of drugs and toxins but rather the modulation of local and systemic metabolism and signaling.^{189,190} Much of this change in our understanding of “what drug transporters really do” is the result of “omics” analyses of knockout mice, human genome-wide association studies (GWAS), and the identification of heritable mutations that cause or modulate well-known metabolic diseases.¹⁸⁴

According to this new systems biology view (explained in more detail later), the multispecificity of these transporters for

endogenous substrates enables a range of drugs and toxins to coopt these transporters expressed in the gut, liver, kidney, and many other tissues. But the pharmaceutical and commercial relevance of these transporters can create the misconception that these widely expressed and evolutionarily conserved genes primarily exist to handle synthetic drugs. Although this is apparently their key role from the perspective of clinical pharmacology and pharmacokinetics, it is becoming clear that this lens is extremely limited even from the clinical point of view, because it is now evident that OAT1, OAT3, and other “drug” transporters are central to endogenous physiology and are important for understanding the pathophysiology of uremia, hyperuricemia, and a number of genetic conditions.

OAT1 KNOCKOUT AND OAT3 KNOCKOUT MICE

Our understanding of in vivo OAT function has changed with the application of omics (e.g., metabolomics, transcriptomics) analyses to *Oat* knockout mice. The *Oat1* and *Oat3* knockout mice were first analyzed over a decade ago and have continued to yield considerable insight into the in vivo function of these two major organic anion transporters, particularly their endogenous function.^{191–196}

As expected, the *Oat1* knockout mouse tissue has defective uptake of the classic organic anion transporter probe PAH, whereas the *Oat3* knockout mouse tissue has defective uptake of estrone sulfate.^{191,197} Consistent with in vitro data, the *Oat1* and/or *Oat3* knockout mice have altered in vivo or ex vivo (e.g., embryonic kidney organ cultures) handling of diuretics

Table 8.1 Some SLC22 Transporter Substrates

Substrate	SLC22 Transporter			
	SLC22A6 OAT1	SLC22A8 OAT3	SLC22A1 OCT1	SLC22A2 OCT2
Nonsteroidal antiinflammatory drugs	✓	✓		
Ibuprofen				
Naproxen				
Antivirals	✓	✓		
Tenofovir				
Adefovir				
Cidofovir				
β-lactam antibiotics	✓	✓		
Ampicillin				
Benzylpenicillin				
Diuretics	✓	✓		
Bumetanide				
Furosemide				
TCA cycle intermediates	✓	✓		
Short-chain fatty acids	✓	✓		
Bile acids		✓		
Flavonoids	✓	✓		
Gut microbiome products	✓	✓	✓	✓
Organic mercurials	✓	✓		
Cisplatin			✓	✓
Metformin			✓	✓
Cimetidine		✓	✓	✓
Thiamine			✓	✓

TCA, Tricarboxylic acid.

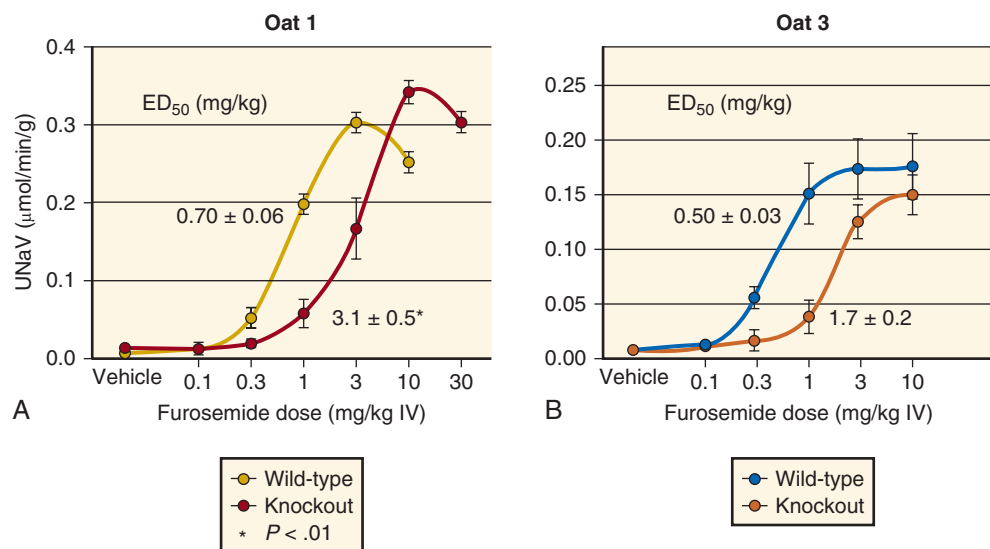


Fig. 8.9 Marked attenuation of thiazide and loop diuretic effects in *Oat1* and *Oat3* knockout mice. In *Oat1* or *Oat3* knockout, which transport diuretics (Table 8.1), luminal sodium elimination is markedly attenuated. ED₅₀, Half-maximal effective dose; IV, intravenous; UNaV, urinary sodium excretion. (Modified from Eraly SA, Vallon V, Vaughn DA, et al. Decreased renal organic anion secretion and plasma accumulation of endogenous organic anions in OAT1 knock-out mice. *J Biol Chem* 2006;281:5072–5083; and Vallon V, Rieg T, Ahn SY, et al. Overlapping in vitro and in vivo specificities of the organic anion transporters OAT1 and OAT3 for loop and thiazide diuretics. *Am J Physiol Renal Physiol*. 2008;294:F867–F873.)

(e.g., loop, thiazides), antibiotics (e.g., penicillin, ciprofloxacin), a wide range of antiviral agents, and methotrexate.^{198–204}

Knockout mice responses to loop and thiazide diuretics provide a useful illustration (Fig. 8.9). These albumin-bound drugs in the peritubular capillaries must be transported by

basolateral uptake transporters OAT1 and OAT3 in the proximal tubule, transit the cell, and exit through apical transporters (including members of the ABCC or MRP families)—all before flowing down the luminal (urinary) space to be excreted or, for the loop and thiazide diuretics, to inhibit

salt reabsorption in later nephron segments. Three to five times more diuretic is required to achieve the same degree of natriuresis after deletion of either *Oat1* or *Oat3* in mice.^{191,198}

OATs are implicated in renal organic mercurial toxicity because mercury binds to glutathione and other thiol-containing compounds, many of which are “effectively” seen as organic anions by the transporter.²⁰⁵ When the *Oat1* knockout mouse was treated with high-dose mercury, the kidneys were surprisingly well protected from injury (histologically and by renal indices), consistent with the inability of the organic mercurial to be taken up by the proximal tubule due to the absence of *Oat1*.²⁰⁶

Perhaps most interesting, and somewhat unexpected, have been the results from performing metabolomics analyses on the plasma of the *Oat* knockout animals^{191,193–195,207} (Fig. 8.10). These have revealed a somewhat surprising range of endogenous OAT substrates.

Generally speaking, both OAT1 and OAT3 appear to play a key role in regulating the flow of organic anions through the so-called gut–liver–kidney axis.¹⁹⁴ This includes the renal handling of many compounds derived from the gut microbiome—either products of the gut microbiome or due to the action of the gut microbiome on dietary components such as phytochemicals. For example, in the *Oat3* knockout, among the greatest changes are in flavonoids that have been acted upon by phase 2 liver enzymes (e.g., glucuronidation). This highlights the important connection between OATs (especially OAT3) with liver metabolism (via so-called drug-metabolizing enzymes) of endogenous compounds as well as drugs and toxins.

Other groups of metabolites elevated in the *Oat3* knockout include primary and secondary bile acids.¹⁹⁴ In *Oat1* and *Oat3* knockouts, fatty acids, TCA cycle intermediates, and vitamins are also elevated. Nevertheless, unlike the case of FDA-approved drugs—where it is difficult to distinguish

molecular properties of drug substrates that predispose to interaction with OAT1 or OAT3²⁰⁸—the sets of metabolites appear fairly distinct, although there is still much overlap.¹⁹⁴

These chemical properties of drugs and metabolites interacting with OAT1 versus OAT3 deserve further mention. With respect to drugs, OAT1 and OAT3 strongly favor anionic drug substrates, but both transporters (especially OAT3) can bind a limited number of cationic/zwitterionic drugs.¹⁸⁰ Cimetidine is an example.¹⁷⁹ Importantly, chemical properties of metabolites interacting with the OATs appear substantially different from the drug data. This could partly be due to a selection factor related to which drugs make it to market. However, it remains that, more than for drugs, the molecular and chemical properties of metabolites appear to distinguish, in a general way, between OAT1 and OAT3 substrates—with OAT3 substrates being larger, less polar, more chemically complex, and containing more ring structures.¹⁹⁴ Nevertheless, it is worth mentioning again that there is overlap between OAT1 and OAT3 metabolite substrates.

In addition, metabolic reconstructions based on changes in gene expression in the knockouts have been performed.^{207–209} These metabolic reconstructions are generally consistent with metabolomics data; the reconstructions indicate that OATs regulate many systemic biochemical pathways as well as those in the proximal tubule (Table 8.2). For example, among the top biochemical pathways revealed in reconstructions are purine metabolism, TCA cycle, fatty acid metabolism, eicosanoid metabolism, amino acid metabolism (tryptophan, tyrosine, arginine), and a variety of vitamin-dependent pathways. Together, the reconstructions and the metabolomics data support the view that OATs are not simply “drug” transporters but impact many aspects of systemic and proximal tubule physiology.

Together, the metabolic reconstructions based on knockout omics data and the analysis of endogenous substrates (e.g.,

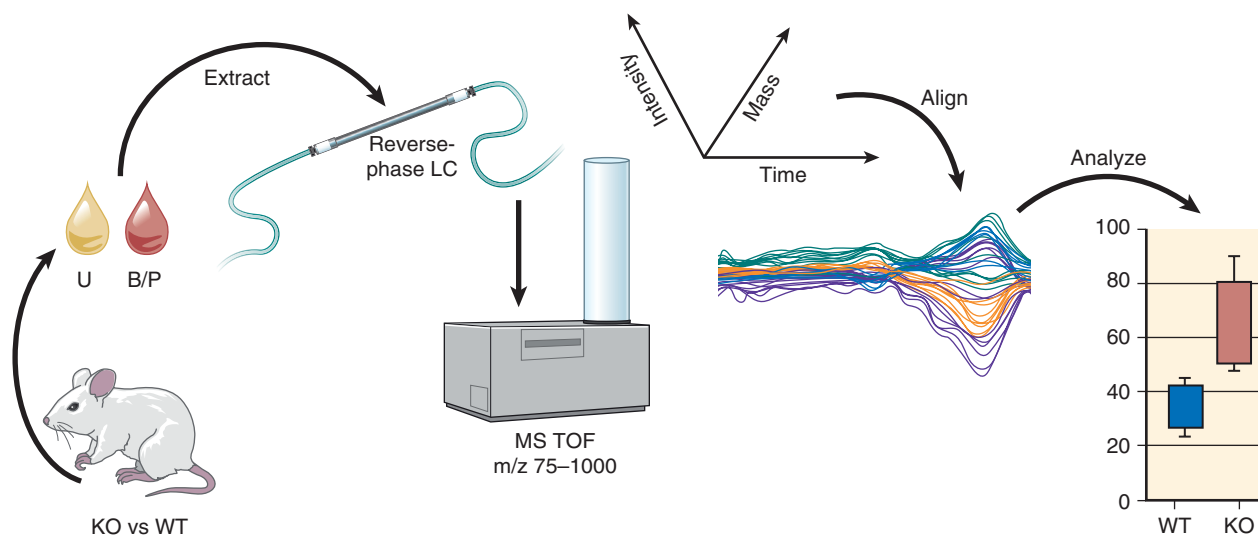


Fig. 8.10 Schematic of approach to metabolomic analysis of *Oat1* knockout mice. Schematic of strategy for untargeted metabolomic analysis of metabolites, gut microbiome (enterobiome) products, vitamins, and signaling molecules, many of which have been shown to directly interact with *Oat1* as determined by *in vitro* assays. Wild-type and *Oat1*^{−/−} mice plasma and urine were profiled using liquid chromatography with tandem mass spectrometry (LC/MS/MS). Importantly, untargeted metabolomics identifies gut microbiome and uremic toxins 1 (*Oat1*). Subsequently, *Oat3* knockouts were also profiled; gut microbiome products and uremic toxins were also identified, as well as a number of bile acids and flavonoid metabolites. *KO*, Knockout; *WT*, wild-type. (Modified from Wikoff WR, Nagle MA, Kouznetsova VL, et al. Untargeted metabolomics identifies enterobiome metabolites and putative uremic toxins as substrates of organic anion transporter 1 (*Oat1*). *J Proteome Res*. 2011;10:2842–2851.)

Table 8.2 Top Pathways Affected by Oat1 Loss in Knockout Mice

TCA cycle
 Tyrosine metabolism
 Alanine, aspartate, and glutamate metabolism
 Butanoate metabolism
 Arginine and proline metabolism
 Tryptophan metabolism
 Nicotinate and nicotinamide metabolism
 Valine, leucine, and isoleucine degradation
 Nitrogen metabolism
 Glyoxylate and dicarboxylate metabolism
 Propanoate metabolism
 Glycine, serine, and threonine metabolism
 Purine metabolism
 Pyrimidine metabolism

TCA, Tricarboxylic acid.

Adapted from Liu HC, Jamshidi N, Chen Y, et al. An organic anion transporter 1 (OAT1)-centered metabolic network. *J Biol Chem*. 2016;291:19474-19486.

metabolites, signaling molecules) from the perspective of chemical properties calls into question the oft-discussed “redundancy” of OAT1 and OAT3 in the proximal tubule. From a practical pharmacokinetic perspective, this view may still be a useful first approximation for many drugs that can interact with both OAT1 and OAT3. But considering the endogenous metabolite preferences alone (without considering drugs), a more appropriate view might be that the two transporters have distinct roles in many metabolic processes, although they work together to handle certain substrates like uric acid. Based on current data, OAT1 appears more linked to local and systemic aerobic metabolism, whereas OAT3 appears more linked to flow of metabolites that originate in the gut or liver (e.g., primary and secondary bile acids). There is also some evidence to suggest that OAT3 could modify phenotypes, for instance, in diabetic disease,²¹⁰ blood pressure,¹⁹² and in the setting of treatment with SGLT2 inhibitors.¹¹⁵

OAT1 AND OAT3 SINGLE-NUCLEOTIDE POLYMORPHISMS

Nonsynonymous coding region single-nucleotide polymorphisms (SNPs) in the OATs are uncommon compared with noncoding region SNPs.²¹¹ Although SNPs in OATs have received less attention than OCT SNPs—because OCTs appear to be more polymorphic in humans—associations have, in recent years, been reported that affect diuretic responsiveness, mercury toxicity, antibiotic levels, and hyperuricemia.^{212–216} A noncoding SNP in the OAT1 gene appears to be associated with the progression of renal disease; whether or not this is related to altered handling of one or more uremic or other toxins is not clear.²¹⁷ Based on animal and human data, it would not be surprising if OAT3 SNPs are found to be associated with glucose homeostasis or diabetic renal disease. Because there is considerable overlap in drugs transported by OAT1 and OAT3, it may be that SNPs in both OATs (or in an OAT and the “corresponding” apical transporter such as MRP2 or MRP4) are required for pronounced drug, toxin, and metabolite phenotypes.²¹⁸ This question needs to be examined in more detail.

OCT2 (SLC22A2)

Unlike the OAT subclade (of the SLC22 OAT major clade), which is large, the OCT subclade (one of three subclades of the SLC22 OCT major clade) consists of three highly homologous (both protein sequence and function) transporters: OCT1, OCT2, and OCT3.¹⁸³ OCTs are generally held to be electrogenic uniporters.²¹⁹ Organic cation transporter 2 (OCT2) is the main renal uptake transporter on the basolateral membrane (blood side) of the proximal tubule cell that is involved in the elimination of organic cationic drugs such as metformin and *cis*-platinum.^{220,221} On the luminal (apical, urine side) of the cell, it appears that MATE (SLC47) transporter family members efflux organic cations into the urinary space.²²² Recently, the primary liver OCT, OCT1—one of the drug transporters that, along with OAT1 and OAT3, has been highlighted by regulatory agencies as important for new drug testing to identify transport mechanisms—has been shown to be a thiamine transporter.²²³

Thus, as with the OATs, OCTs may function primarily in regulating metabolite flow into and out of tissues, and as with OATs, the best in vivo functional information regarding endogenous function comes from analysis of metabolites altered in the knockout mice. On the whole, the known drug and metabolite substrates for the OCTs appear less diverse than for the OATs, and it appears that, based on molecular property analysis of drug substrates, OCT1 and OCT2 have largely overlapping specificities, at least for drugs.¹⁸⁰

OCT2 SINGLE NUCLEOTIDE POLYMORPHISMS

The OCTs appear more polymorphic than the OATs; SNPs in OCT2 have received considerable clinical attention because, consistent with in vitro studies, they can affect levels of the antidiabetic agent metformin and the chemotherapeutic agent *cis*-platinum.^{224–227} These associations are much better established than the SNP associations mentioned above for the OATs.²²⁸ Because metformin and *cis*-platinum have been so widely used, and because of the potential for toxicity, it is important for the clinician to be aware of the possibility that SNPs in OCT2 can affect drug levels.

Clinical Relevance

The Remote Sensing and Signaling Hypothesis: A Framework for Understanding Hyperuricemia and Uremia

The OATs and OCTs have recently garnered a great deal of attention because of regulatory concerns due to transporter-mediated drug–drug interactions. Much more is likely on the way regarding drug-metabolite interactions (DMI) with the advent of better techniques to analyze metabolites in tissues and body fluids. With new approaches to uremia and hyperuricemia (especially in CKD) being considered, one expects more clinical studies and trials aimed at decreasing the burden of uremic toxins and uric acid—and prolonging time to severe CKD and dialysis. In this regard, the Remote Sensing and Signaling Hypothesis should be useful for considering approaches to ameliorate perturbed inter-organ and interorganismal communication via metabolites and signaling molecules, including those derived from the gut microbiome.

APICAL MEMBRANE PROXIMAL TUBULE TRANSPORTERS INVOLVED IN THE HANDLING OF ORGANIC ANIONS, ORGANIC CATIONS, AND ORGANIC ZWITTERIONS

In the proximal tubule, the OAT1, OAT3, and OCT2 function in basolateral side uptake (influx) of organic anions and cations from the blood. Although what happens to these charged organic molecules inside the cell remains poorly defined, their exit, usually in unchanged form, has become better understood in recent years. Many of the organic anions taken up by OAT1 and OAT3, for example, are effluxed across the apical membrane by members of the ABCC family, MRP2 (ABCC2) and MRP4 (ABCC4).¹⁶⁸ These are not the only apical transporters of organic anions in the proximal tubule; for example, OAT4 and the well-known ABC transporters, P-glycoprotein and ABCG2, appear to play a role for certain substrates.¹⁶⁸ Although in simplified representations, there is a tendency to match basolateral OAT1 and OAT3 with apical MRP2 and MRP4, it is likely that, depending on the anionic substrate taken up by OAT1 and OAT3, one or several apical transporters are involved in apical efflux into the proximal tubule lumen.

With respect to the apical efflux of organic cations, in recent years, MATEs, particularly MATE-2K, have received attention.^{168,222} MATEs are members of the SLC47 family, and it is well established that they transport many cationic drugs taken up by OCT2. Other apical transporters might also transport some OCT2 substrates, but their contribution is not well defined. It is also important to mention the organic cation/zwitterion transporters OCTN1 (SLC22A4) and OCTN2 (SLC22A5). Although they are sometimes listed along with drug transporters, their endogenous substrates—ergothioneine and carnitine, respectively—are well established.^{229,230} Indeed, mutations in OCTN2 cause systemic carnitine deficiency, which can lead to severe cardiac and skeletal myopathy. Ergothioneine, on the other hand, is considered an important antioxidant. Nevertheless, OCTNs have some ability to interact with cation drugs.

RENAL TRANSPORT OF SPECIFIC ORGANIC SUBSTRATES IN DISEASE

UREMIC TOXIN TRANSPORT

The importance of OATs in renal handling of gut microbiome-derived metabolites merits further discussion. Many of the metabolites accumulating in the *Oat* knockouts, such as indoxyl sulfate, kynurenine, p-cresol sulfate, and hippurate, are among the sets of gut microbiome-derived small molecules (organic anions); they are also frequently implicated as “uremic toxins”^{193,195} (Table 8.3).

The list of small molecules implicated in uremic toxicity is long and much debated.^{231,232} These uremic toxins are thought to play a role in many tissue and organ toxicities and dysfunctions that occur in the uremic syndrome associated with severe CKD. It is unlikely that any single uremic toxin on the list is the key to all the manifestations of the uremic syndrome, although there is growing evidence that certain uremic toxins play a role in particular tissue toxicities. These include TMAO (trimethylamine-N-oxide), which has been implicated in cardiovascular toxicity,^{233,234} and indoxyl sulfate, which has been implicated in multiple aspects of the uremic syndrome.²³⁵

Table 8.3 Uremic Toxins Accumulating in *Oat1* and/or *Oat3* Knockout Mice

Indoxyl sulfate	Indolelactate
p-Cresol sulfate	Kynurenate
Hippurate	Putrescine
CMPF	Uric acid
Phenyl sulfate	Creatinine
Xanthurenate	

CMPF, 3-Carboxy-4-methyl-5-propyl-2-furanpropanoate.

Adapted from Wikoff WR, Nagle MA, Kouznetsova VL, et al.

Untargeted metabolomics identifies enterobiome metabolites and putative uremic toxins as substrates of organic anion transporter 1 (*Oat1*). *J Proteome Res.* 2011;10:2842-2851; and Wu W, Bush KT, Nigam SK. Key role for the organic anion transporters, OAT1 and OAT3, in the in vivo handling of uremic toxins and solutes. *Sci Rep.* 2017;7:4939.

Some of these uremic toxins (e.g., indoxyl sulfate) may play a role in the actual progression of renal disease, presumably via OAT-mediated uptake into proximal tubule cells.^{195,236} The pathophysiology of uremia is beyond the scope of this chapter, and readers are referred instead to Chapter 52. But it is important to emphasize the growing appreciation of the role of OAT1 and OAT3 in regulating levels of many molecules considered uremic toxins. Thus, it will be interesting to determine whether SNPs, or other factors that affect the expression or function of OAT1 and OAT3, alter the time at which the uremic syndrome develops or the progression of renal disease.

OCT2 also appears to be a transporter of certain cationic uremic toxins, notably TMAO, a molecule accumulating in CKD that is associated with cardiovascular disease.²³⁷ The apical transporters of organic anions, the MATEs, appear to be involved in the efflux of TMAO. Of note, the deletion in mice of OAT3, which can transport a few organic cations despite being an organic anion transporter, results in elevated levels of TMAO, although it is not clear whether TMAO is actually transported by OAT3.¹⁹⁵

URIC ACID TRANSPORT

Uric acid is an antioxidant that also has deleterious effects.²³⁸ It is also on the list of “uremic toxins.” Apart from gout and kidney stones, hyperuricemia has been associated with cardiovascular disease, metabolic syndrome, hypertension, and progression of renal disease.²³⁹ Metabolism of purines by enzymes primarily in the liver (e.g., xanthine oxidase) results in uric acid formation. On the other hand, uric acid elimination and retention in the body is largely determined by the kidney and, to a lesser extent (in the absence of kidney disease), by the intestine.

One of the unexpected results of GWAS and knockout mouse studies from the perspective of uric acid homeostasis was the number of SLC and ABC “drug” transporters that were implicated in regulating serum uric acid levels and then shown to transport urate.^{238,240} These transporters include ABCG2 (also known as BCRP), URAT1 (a close relative of OAT1 and OAT3), OAT1 and OAT3, as well as other OATs. In murine knockouts of OAT1, OAT3, and URAT1 (SLC22A12, originally discovered as *Rst* in mice²⁴¹), there are alterations in renal urate handling, although they are not as great as

might have been expected if these genes were, as was thought by many at the time, the major contributors to renal urate handling.¹⁹⁶ Subsequently, another SLC transporter, related to the glucose transporters, known as GLUT9 (SLC2A9), was found to be very important for renal urate handling.²⁴² Other transporters implicated in uric acid transport are MRP2 (ABCC2), MRP4 (ABCC4), NPT1 (SLC17A1), NPT4 (SLC17A3), OAT4 (SLC22A11), and OAT10 (SLC22A13).²⁴³ The list of associated genes, which includes nontransporters as well, continues to grow as more analyses are done in different ethnic groups. Indeed, based on GWAS and other studies, it appears that different subsets of the aforementioned genes may be more or less important in urate handling, depending on ethnicity and gender.

Together, these transporters are responsible for the complex handling of uric acid by the kidney, but it is important to emphasize that ABCG2 is increasingly perceived as the main intestinal uric acid efflux transporter.²⁴⁴ Intestinal extrusion of urate becomes particularly important in the setting of severe renal insufficiency, and in this regard, it has been found that SNPs in ABCG2 become more highly associated with uric acid levels in CKD patients²¹⁵ (Fig. 8.11). This seems

to be an example of remote organ communication (i.e., between injured kidney to intestine; see also later) with the apparent “objective” of optimizing uric acid levels as transport in another organ (intestinal efflux) takes over from the declining kidney, thereby reducing the plasma uric acid levels. From studies in animal models, it is believed that the high uric acid level itself in CKD leads to increased expression and/or function of intestinal ABCG2, and if so, this may be a case of substrate induction of the transporter.²⁴⁵

CREATININE TRANSPORT

Although new measures may soon become routinely used in hospitals, serum creatinine continues to be considered an index of renal function both in primary care and hospital settings. A number of members of the SLC22 family (containing OATs and OCTs) transport creatinine. These include OCT2, OAT2, OAT3, and possibly OAT1.^{246–248} All of these are considered key transporters of organic cation drugs, organic anion drugs, and zwitterionic drugs. Their relative importance in renal creatinine handling is debated, but it is likely that all play some role in the renal handling of creatinine, with OCT2 and OAT2 being particularly important, at least based on more recent

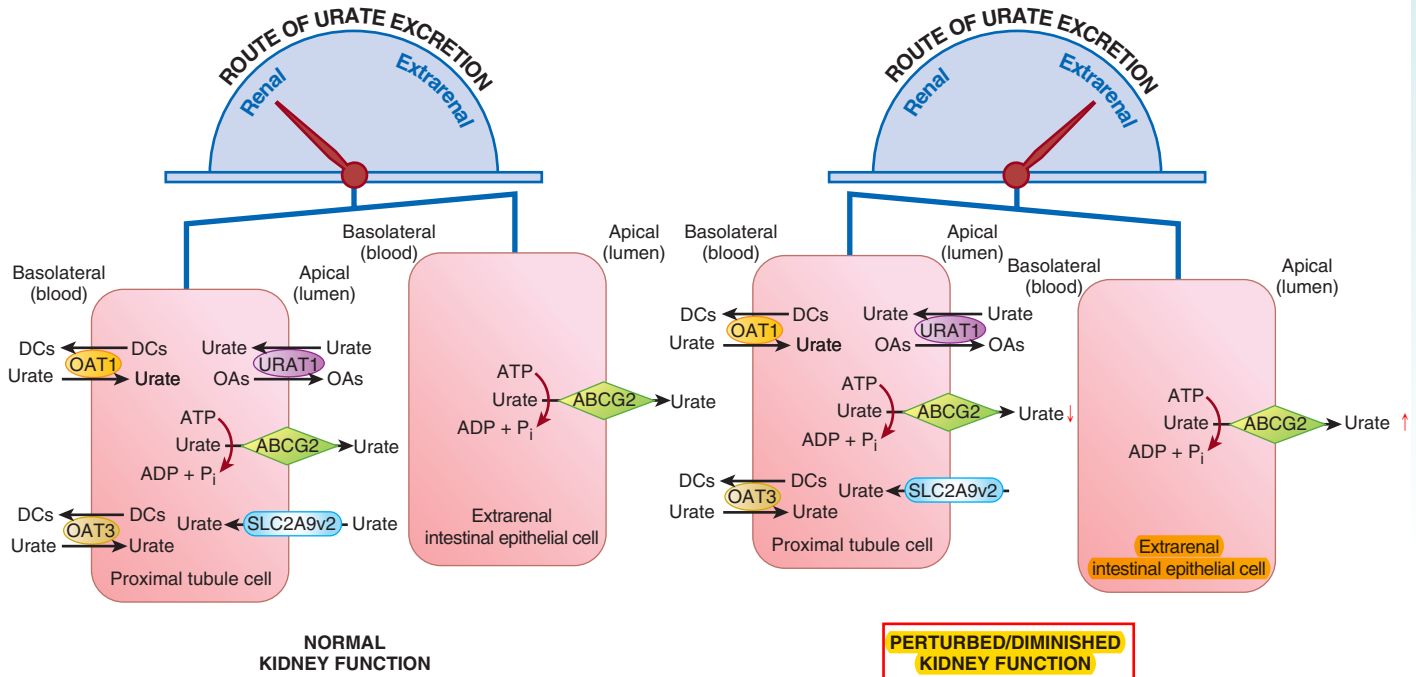


Fig. 8.11 In chronic kidney disease, the intestinal role of uric acid transporters (mainly ABCG2) becomes much more important.

Schematic of urate excretion in the setting of normal renal function versus that in which there is diminished renal function. In the normal situation, the majority of urate excretion (~70%) is performed mainly in the proximal tubule of the kidney and is mediated by a number of transporters found on the apical and basolateral membranes of the proximal tubule cell. The illustration depicts only some of the transporters involved and includes members of the SLC and ABC families of membrane transporters. On the basolateral surface, the uptake of uric acid from the blood is mediated mainly by OAT1 (SLC22A6) and OAT3 (SLC22A8). These solute carrier transporters exchange dicarboxylates (DCs) for urate resulting in the net movement of this organic anion into the proximal tubular cell. At the apical surface, a number of transporters, including ABCG2 (depicted here), NPT1 (SLC17A1), and NPT4 (SLC17A3), as well as ABCC family members, work to secrete urate into the tubular lumen for excretion via the urine. The relative importance of the aforementioned apical transporters in urate secretion remains unclear. Transport of urate via ABCG2 is driven through ATP hydrolysis. Several other apical membrane transporters are well established in the reabsorption of urate, including URAT1 and SLC2A9v2. URAT1, or SLC22A12, exchanges intercellular organic anions for urate, resulting in the movement of uric acid back into the proximal tubule cell. Meanwhile, under normal physiologic conditions, up to 30% of uric acid is excreted via extrarenal transporters, believed to largely be driven by ABCG2 expressed in intestinal epithelial cells. The results of the analysis of human data—as well as physiologic data from rodent models with renal dysfunction—support the view that urate transport by ABCG2, most likely in intestine, compensates for poor renal urate handling in the setting of diminished kidney function. (Modified from Bhatnagar V, Richard EL, Wu W, et al. Analysis of ABCG2 and other urate transporters in uric acid homeostasis in chronic kidney disease: potential role of remote sensing and signaling. *Clin Kidney J.* 2016;9:444–453.)

studies. The issue is clinically relevant, because a number of drugs (e.g., trimethoprim) are thought to “artificially” create the impression of renal dysfunction when creatinine is the primary measure used because of drug–metabolite interactions at the level of the transporter.¹⁶⁸

THE REMOTE SENSING AND SIGNALING HYPOTHESIS: A FRAMEWORK FOR UNDERSTANDING HYPERURICEMIA AND UREMIA

It is evident from in vitro transport studies, metabolomics analyses of knockout animals, GWAS studies, and metabolic diseases due to transporter mutations that multispecific “drug” transporters of both the SLC and ABC transporter family are critical in the local and systemic regulation of levels of a huge array of metabolites.^{183,184,189,190,249}

The Remote Sensing and Signaling Hypothesis—which began to be formulated in 2004–2007¹⁸⁶—argues that the physiologic role of multispecific “drug” transporters and their close (often mono- or oligo-specific) relatives is the regulation of “remote” interorgan and interorganismal communication via metabolites and signaling molecules by these SLC and ABC transporters (Fig. 8.12).

The system is hypothesized to be actively regulated through transcriptional mechanisms (e.g., nuclear receptors) and posttranslational mechanisms (e.g., kinases regulating

transporter internalization or PDZ domain association). It is envisioned to work in parallel with the neuroendocrine, growth factor cytokine, and autonomic systems. The argument is that this type of SLC and ABC transporter-mediated remote interorgan and remote interorganismal communication via small molecules of “high informational content” (e.g., rate-limiting metabolites, signaling molecules, antioxidants, vitamins) is as critical as these other homeostatic systems and, as such, deserves comparable consideration for understanding health and disease states.

The hypothesis seems to be particularly relevant to renal disease. Consider, for instance, the previously discussed example of apparent interorgan remote communication—in order to reestablish uric acid homeostasis.²⁴³ An injured organ—the malfunctioning kidney in CKD—is involved in a kind of “organ cross-talk” with the intestine, whereby intestinal ABCG2 transporters, and possibly others, appear to “take over” from many renal SLC uric acid transporters as tubular function declines. As discussed earlier, high urate, certain uremic toxins, or both—accumulating as a consequence of renal disease—are thought to induce expression of intestinal transporters.

More generally, considering the uremic syndrome as partly due to “pathologic” transporter-mediated remote interorgan and interorganismal communication of uremic toxins—which includes signaling molecules that bind G-protein-coupled

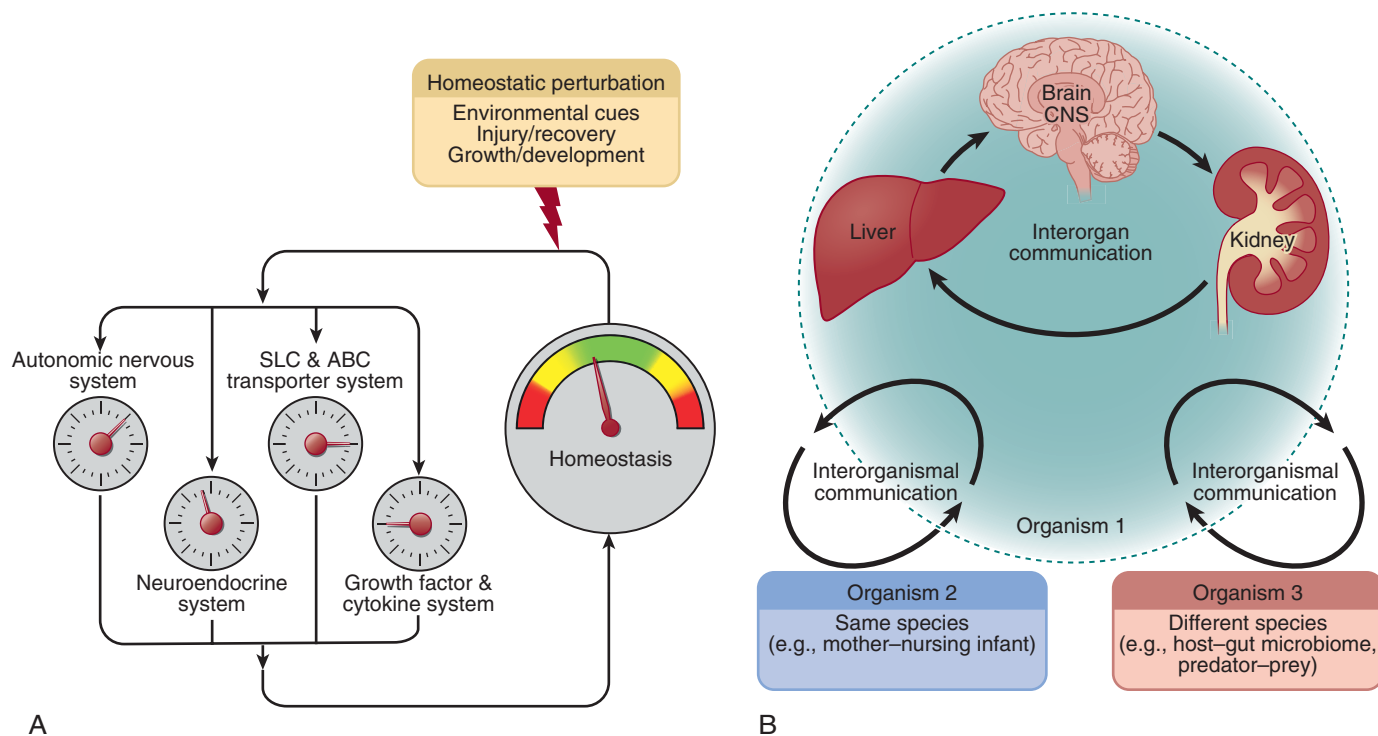


Fig. 8.12 The Remote Sensing and Signaling Hypothesis as it relates to multispecific and more selective SLC and ABC transporters as well as drug-metabolizing enzymes (DMEs) in different tissues communicating “remotely” (interorgan and interorganismal) in part due to differentially expressed transporters via the gut–liver–kidney axis in health and disease. The Remote Sensing and Signaling Hypothesis emphasizes the role of multispecific (“drug” transporters like OAT1, OAT3, ABCG2, and MRPs) and more selective SLC22 transporters as well other SLC and ABC drug and DMEs in a remote interorgan and interorganism communication network involving transporters and DMEs in epithelial and nonepithelial tissues as well as various body fluid spaces, such as cerebrospinal fluid, breast milk, and urine. This remote sensing and signaling system works in concert with more classic systems involved in homeostasis and resetting homeostasis in the setting of diseases like hyperuricemia and uremia. These other systems include the neuroendocrine, growth factor cytokine, autonomic nervous systems, and increasing evidence indicates that it is intertwined with these classic homeostatic systems. CNS, Central nervous system. (Adapted from Nigam SK. What do drug transporters really do? *Nat Rev Drug Discov.* 2015;14:29–44.)

receptors (e.g., kynurenine) and nuclear receptors (e.g., indoxyl sulfate)—provides a different lens on the numerous biochemical and cellular aberrations found in the uremic syndrome, which affect many tissues and body compartments. Moreover, the source of many uremic toxins in the body is the gut microbiome—the result of interorganismal communication—and these toxins affect signaling and metabolism in multiple organs before, in many cases, being eliminated by proximal tubule multispecific “drug” transporters, such as the OAT1, OAT3, OCT2, and the MATEs.

The Remote Sensing and Signaling Hypothesis provides a systems pathophysiology framework for thinking about the many metabolic and signaling aberrations of the uremic syndrome in the context of remote interorgan and interorganismal small molecule communication. Such a systems-level view may lead to the consideration of new therapeutic approaches aimed at altering remote sensing and signaling mechanisms as they become better understood—with the hope of ameliorating some of the many harmful manifestations of the uremic syndrome.

DRUG–METABOLITE INTERACTIONS

As more is learned about the endogenous substrates of OATs, OCTs, and other drug transporters, and as more is learned through metabolomics and metabolic reconstructions (such as those described above) regarding the regulation of tissue-specific metabolism by these transporters, it will become possible to consider new ways to modulate complex metabolic diseases like uremia and to understand the implications of drug–metabolite interactions (DMI) beyond simple competition at the level of the transporter itself.^{168,208} For example, currently the thinking on DMI is largely limited to looking at the transporter-level competition between the drug trimethoprim, which binds OCT2 and thereby raises the plasma concentration of a single OCT2-transported metabolite such as creatinine. But in the case of OAT1, it is now well established from knockout studies that loss of OAT1 function affects many metabolic pathways.^{208,209} The implication is that a drug that competes for OAT1 binding with endogenous metabolites normally transported by OAT1 (e.g., probenecid) will have broad effects on metabolism, affecting metabolites and signaling molecules that may themselves not be direct OAT1 substrates. This type of DMI may help partly explain the broad metabolic syndrome-like effects seen in the setting of chronic use of certain OAT-transported drugs, such as diuretics and HIV antivirals.^{250,251} Much work needs to be done in this area of DMI where competition for transport by the drug may affect a wide range of metabolic pathways; this would seem particularly relevant in the setting of moderate CKD where the affected metabolites would likely include certain uremic toxins.¹⁶⁸

AMINO ACIDS

PHYSIOLOGY OF RENAL AMINO ACID TRANSPORT

Rare inherited defects of renal amino acid transport have been instrumental for our understanding of renal metabolite reabsorption.^{252,253} Four disorders associated with apical amino

acid transporters define four major renal transport pathways, namely, a transporter for neutral amino acids (mutated in Hartnup disorder), a transporter for cationic amino acids and cystine (mutated in cystinuria), a transporter for anionic amino acids (mutated in dicarboxylic aminoaciduria), and a transporter for glycine and proline (mutated in iminoglycinuria)²⁵⁴ (see Chapter 44 for a detailed discussion of these inherited disorders). With the exception of iminoglycinuria, aminoacidurias also affect intestinal transport. Genetic complexity is observed in cystinuria and iminoglycinuria, demonstrating the involvement of more than one gene in the transport process. Apart from intestinal transport, aminoacidurias affecting apical transport have little effect on other organs, indicating that expression of these transporters is largely specific to the apical membrane in the kidney and intestine, although some renal amino acid transporters are also found in the brain. The four major amino acid transport activities were verified further by *in vitro* studies using a variety of methods such as microperfusion studies, brush border membrane vesicles, cortical slices, and cell lines.^{254,255} *In vitro* studies, in addition, identified a fifth transport activity, namely for β -amino acids, for which no corresponding disorder exists. Transport across the basolateral membrane has been more challenging to delineate, and it is unclear whether the five apical pathways are matched by five basolateral exit routes. A clear pathway for the release of cationic amino acids has been genetically identified through the rare amino aciduria lysinuric protein intolerance (LPI) and through functional studies.^{256,257} A release pathway for neutral amino acids was defined functionally using vesicles derived from the basolateral membrane.²⁵⁸ Release of anionic amino acids is difficult to measure, due to significant metabolism of glutamate by epithelial cells²⁵⁹ (see also Fig. 8.4). Glycine is likely to join other neutral amino acids for efflux, but the efflux pathways for proline and β -amino acids remain unclear. More than 98% of all filtered amino acids are reabsorbed in the proximal tubule; other parts of the tubule thus do not significantly contribute to amino acid reabsorption and are not discussed here.²⁶⁰ Some differences in transporter expression are observed between the PCT and the PST; these are illustrated in Fig. 8.13. Species differences may occur in the kidney and are mentioned where relevant.

MOLECULAR BIOLOGY OF RENAL AMINO ACID TRANSPORTERS

Molecular cloning, human genetics, and mouse models have helped to identify almost all amino acid transporters in the apical and basolateral membrane.^{254,261} Renal epithelial amino acid transporters are found in a variety of SLC families (Table 8.4). The SLC nomenclature is generally used for the genes, whereas acronyms are typically used for the proteins that describe some of the properties of the transporter. Expression cloning using *Xenopus laevis* oocytes or mammalian cell lines has been instrumental in the identification of renal amino acid transporters.²⁶² Additional transporters were identified by sequence similarity. As a result of these efforts, the amino acid transporter endowment of renal epithelial cells is now well understood, and an overview is depicted in Fig. 8.13. In the following, transporters for each group of amino acids are described in detail.

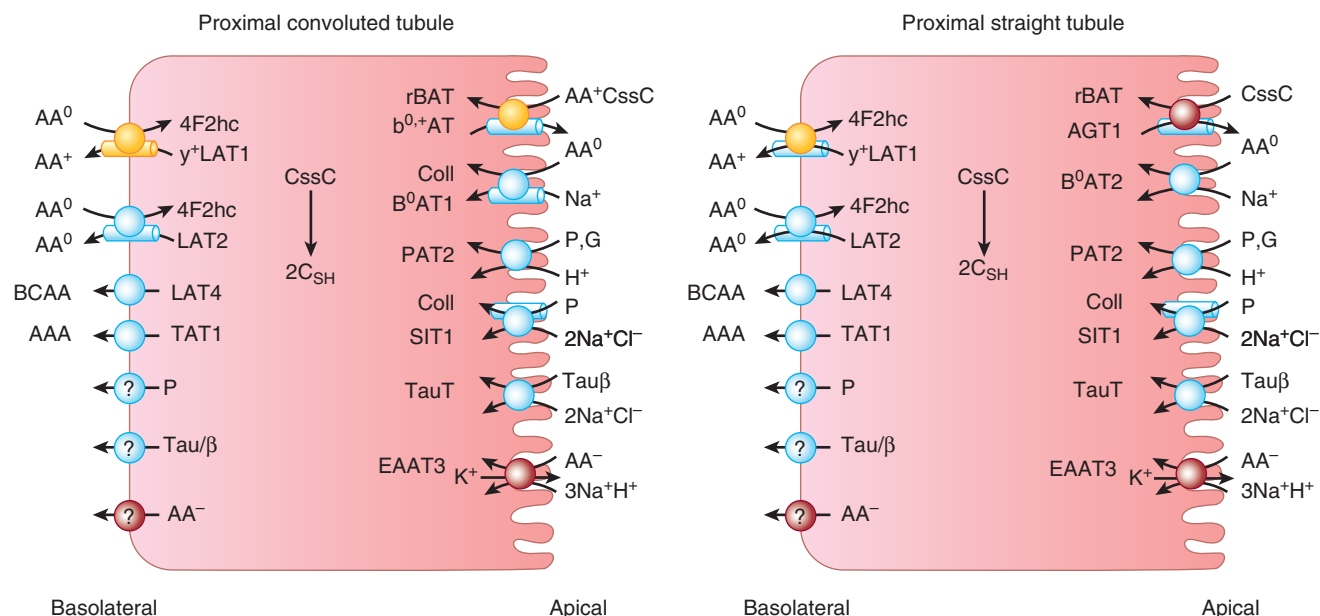


Fig. 8.13 Amino acid transporters in the proximal tubule. The common name of amino acid transporters in the proximal convoluted tubule and proximal straight tubule are shown next to each transporter. Transporter requirement for ancillary subunits are indicated by a tubelike structure. Transporters for anionic amino acids are shown in red; transporters for cationic amino acid are shown in yellow. AA⁰, neutral amino acids; AA⁺, cationic amino acids; AA⁻, anionic amino acids; AAA, aromatic amino acids; BCAA, branched-chain amino acids; C_{SH}, cystine; CysC, cystine; G, glycine; P, proline; Tau/β, taurine and β-amino acids.

Table 8.4 Human Tubular Amino Acid Transporters Properties and Distribution of Human Renal Tubular Transporters

Amino Acid Transporter	SLC	PCT/PST	Substrates	Affinities	Disease	Structure class
Apical						
B ⁰ AT1	SLC6A19	PCT	All neutral	800–15,000 μM	Hartnup disorder (OMIM 234500)	LeuT
B ⁰ AT2	SLC6A15	PCT	BCAA, Met, Pro	50–200 μM	n.r.	LeuT
Collectrin	TMEM27	PCT	N/A	Ancillary	n.r.	1TM
rBAT	SLC3A1	PCT < PST	N/A	Ancillary	Cystinuria (OMIM 220100)	1TM
b ⁰ ,+AT	SLC7A9	PCT > PST	Arg, Lys, Orn, CysC, Met, Leu, Ala	100 μM	Cystinuria (OMIM 220100), isolated cystinuria (OMIM 238200)	LeuT
EAAT3	SLC1A1	PCT < PST	Glu, Asp, CysC	20–80 μM	Dicarboxylic amino aciduria (OMIM 222730)	Git
AGT1	SLC7A13	PCT = PST	Glu, Asp, CysC	20–60 μM	n.r.	LeuT
PAT2	SLC36A2	PCT			Iminoglycinuria (OMIM 242600), Hyperglycinuria (OMIM138500)	LeuT
SIT	SLC36A2	PCT < PST			Iminoglycinuria (modifier) (OMIM 242600)	LeuT
TauT	SLC6A6	PCT, PST			n.r.	LeuT
Basolateral						
LAT2	SLC7A8	PCT			n.r.	LeuT
4F2hc	SLC3A2	PCT = PST		Ancillary	Lethal	1TM
y ⁺ LAT1	SLC7A7	PCT = PST			Lysinuric protein intolerance (OMIM 222700)	LeuT
TAT1	SLC16A10	PCT			n.r.	MFS
LAT4	SLC43A2	PCT = PST	BCAA, Met, Phe	5000 μM	n.r.	MFS

BCAA, Branched-chain amino acids; CysC, cystine; 1TM, single transmembrane-helix protein; MFS, multifacilitator superfamily; n.r., not reported; OMIM, Online Mendelian Inheritance in Man; PCT, proximal convoluted tubule; PST, proximal straight tubule. Links to Diseases Refer to the OMIM Database.

TRANSPORTERS FOR NEUTRAL AMINO ACIDS

APICAL TRANSPORTERS

The presence of a dominant transporter for neutral amino acids can be inferred from the aminoaciduria observed in Hartnup disorder, which is restricted to neutral amino acids, but affecting every member of this group.²⁶³ This transporter was identified as the amino acid transporter B⁰AT1 (Broad neutral (0) Amino acid Transporter 1, SLC6A19).^{264,265} Although Hartnup disorder shows simple recessive inheritance and therefore is monogenic, the transporter protein requires association with ancillary proteins to traffic to the apical membrane and to be fully functional.²⁶⁶ In the kidney, this is facilitated by collectrin (TMEM27),²⁶⁷ whereas in the intestine, this role is served by angiotensin-converting enzyme 2 (ACE2).²⁶⁸ Both proteins are type I transmembrane (TM) proteins with a single TM domain. Although ACE2 is expressed in the kidney, its expression levels in the proximal tubule are too small to make a significant contribution to B⁰AT1 surface expression. B⁰AT1 and TMEM27 are both expressed in the PCT.²⁶⁷ Consistent with the monogenic inheritance, mutations in TMEM27 have not been observed in Hartnup disorder and would be expected to show an amino acid transporter defect in the kidney but not intestine. Thus far, more than 20 different causative mutations have been identified in the *SLC6A19* gene.²⁶⁹ Interestingly, rare variants in *SLC6A19* have been associated with low serum creatinine levels and may have a kidney protective effect or affect creatinine synthesis.²⁷⁰ B⁰AT1 transports all neutral amino acids in symport with 1Na⁺; in contrast to many other members of the SLC6 family, chloride ions are not cotransported.²⁷¹ Substrate affinities range from 1–12 mM, with a preference for branched-chain amino acids (BCAA) and methionine, followed by large hydrophilic and aromatic amino acids.²⁷² In *in vitro* systems proline and tryptophan are very poor substrates of B⁰AT1,

but *in vivo* data suggest a significant contribution to the transport of both amino acids.²⁷³ Mice lacking B⁰AT1 replicate the human aminoaciduria of Hartnup disorder but do not show any additional pathology (Table 8.5). In addition to the low-affinity B⁰AT1 transporter in the PCT, functional studies suggest the presence of a high-affinity transporter for neutral amino acids in the PST. A candidate could be B⁰AT2 (SLC6A15), which is expressed at low levels in the proximal tubule.²⁷⁴ B⁰AT2 has a narrower substrate specificity than B⁰AT1, showing a strong preference for BCAA and methionine with substrate affinities <100 μM. Like B⁰AT1, the transporter is Na⁺ dependent and chloride independent.²⁷⁵

Specific Apical Transporters for Proline and Glycine

Proline and glycine have unusual physicochemical properties, which cause these amino acids to be inefficiently transported by more broadly specific amino acid transporters. Glycine is lacking a side chain, reducing its affinity to side-chain binding pockets, whereas proline has a secondary amino group and restricted flexibility. As a result, proline is a poor substrate for B⁰AT1 and is not recognized by the basolateral neutral amino acid transporter LAT2 (see later). A common transporter for glycine, proline, and hydroxyproline in humans is supported by two lines of evidence. First, in the rare disorder iminoglycinuria, all three amino acids are found in the urine.²⁷³ Second, prolinemia, when passing the renal threshold, causes prolinuria, hydroxyprolinuria, and glycinuria.²⁷⁶ Although iminoglycinuria is an autosomal recessive disorder, it shows clear signs of genetic complexity.²⁷⁷ For instance, some cases of iminoglycinuria show malabsorption of proline in the intestine. Moreover, in some cases, heterozygotes are normal, whereas in other pedigrees, hyperglycinuria is observed. These cases can all be explained by the combined action of proton–amino acid transporter 2 (PAT2, SLC36A2), the system IMINO

Table 8.5 Physiologic Characteristics of Knockout Mice of Renal Amino Acid Transporters

Transporter	Plasma AA	Urine AA	Renal Pathology	Other Features
rBAT	Normal	+++ Lys, Arg, Orn, CysC	Nephritis, kidney stones	None
b ⁰ +AT	+: His, Ser, Glu/n	+++ Lys, Arg, Orn, CysC, + Glu/n	Glomerular fibrosis, nephritis, kidney stones	None
B ⁰ AT1	Normal	+++ Neutral AA	None	Propensity for colitis, improved glycemic control
B ⁰ AT3	Normal	+++ Gly ++ Ala, Val, Leu, Ile, Met, Ser, Thr, Gln, Phe, Tyr	Reduced serum creatinine None	Stress-induced increase of blood pressure
TAT1	+: Phe, Tyr, Trp -: Gly, Ala, Met, Ser, Thr, Asn, Gln HPD +: Phe, Tyr, Trp	+: Phe, Tyr, Trp HPD +: Val, Ile, Leu, Thr, Gln, His, Phe, Tyr, Trp	None	None
LAT4	-: Ala, Pro, His, Ser	Not available	None reported	Liver inflammation, malnutrition
LAT2	+: Gly, Ala, Ser, Thr, Gln, Val, Lys	+: Gly, Ser, Thr, Gln, Leu, Val	None	None
y ⁺ LAT1	Not available	+++ Lys, Arg, Orn	None	Failure to thrive, intrauterine growth restriction
TauT	-: Taurine -: Glu	+++ Taurine	Enlarged kidney, glomerulosclerosis, nephropathy	Muscle weakness, cardiomyopathy, retinal degeneration, hearing loss, chronic liver disease

CysC, Cystine; Glu/n, glutamate or glutamine; HPD, high-protein diet; +, elevated, ++ significantly elevated, +++ highly elevated; -, reduced, --, significantly reduced.

transporter SIT1 (SLC6A20), and the general neutral amino acid transporter B⁰AT1 (SLC6A19).²⁷³ Homozygous mutations in PAT2 account for iminoglycinuria, whereas heterozygous mutations cause selective hyperglycinuria. SIT1 mutations contribute to the iminoglycinuria phenotype and also explain a sporadically observed reduced proline absorption in the intestine. B⁰AT1 provides the baseload for proline and glycine reabsorption, explaining why the extent of aminoaciduria is well below the filtered amounts.²⁷³ The related glycine and alanine transporter B⁰AT3 (SLC6A18) is only functional in mouse^{278,279} (Table 8.5) but not in higher mammalian species, where its function has been replaced by PAT2.

PAT2 (SLC36A2) has been identified as the specific transporter for small neutral amino acids, specifically alanine, glycine, and proline. Single N-methylation of amino acids is tolerated such as in sarcosine or proline. PAT2 is mutated in all cases of iminoglycinuria and is expressed in the proximal tubule of the kidney.²⁷³ The K_M values for its substrates range from 0.1–0.6 mM.²⁸⁰

SIT1 (SLC6A20) accepts only amino acids with secondary, tertiary, or quaternary amines, such as proline, sarcosine, betaine, and methylaminoisobutyric acid.^{281,282} SIT1 is an Na⁺ and Cl[−] dependent transporter, transporting 1 substrate, 2Na⁺ and 1Cl[−] ion. This generates a net positive transporter current, which can be observed when expressed heterologously. Expression of the transporter at the cell surface requires collectrin (TMEM27).²⁸³

B⁰AT3 (SLC6A18) preferentially transports alanine and glycine.²⁷⁹ SLC6A18 knockout mice have hyperglycinuria and slightly elevated urine levels of other neutral amino acids,²⁷⁸ but humans with homozygous mutations in SLC6A18 are normal because even the normal allele is nonfunctional.^{273,284} The mouse transporter requires collectrin coexpression for trafficking and also for its catalytic function.²⁸⁴

Specific Apical Transporters for B-Amino Acids

This group of amino acids consists of taurine, β-alanine, and its homolog gamma-aminobutyric acid (GABA). Plasma concentration of GABA is very low (about 0.1 μM), while taurine levels are significantly higher (about 50 μM). Both amino acids are often accepted by the same transporters, but in the kidney, specific GABA transporters are observed as well. Reabsorption of these amino acids is mainly mediated by the taurine transporter TauT (SLC6A6).²⁸⁵ PAT2 shows weak affinity for these substrates but is probably physiologically irrelevant. TauT translocates taurine with a K_M of 20 μM in a process that involves the cotransport of 2Na⁺ and 1Cl[−]. Urinary taurine excretion in TauT-deficient mice reaches the filtered amounts.²⁸⁶

BASOLATERAL TRANSPORTERS

Functional studies identified an Na⁺-independent transporter broadly specific for neutral amino acids also in the basolateral membrane. The heteromeric amino acid transporter 4F2hc-LAT2 (SLC3A2-SLC7A8) matches with regard to the substrate specificity but mechanistically is an antiporter.^{287–289} A uniporter has been identified for aromatic amino acids (TAT1, SLC16A10),²⁹⁰ but this transporter cannot explain the net flux of amino acids across the renal epithelium as illustrated by *Tat1* knockout mice.²⁹¹ These mice show elevated levels of aromatic amino acids in the urine but normal levels of other amino acids. Only on a high-protein diet, a more general

aminoaciduria was observed, suggesting cooperation between LAT2 and TAT1. LAT4 (SLC43A2) allows efflux of BCAA, methionine, and phenylalanine, thus covering large neutral amino acids.²⁹² A specific efflux pathway for small neutral amino acids has not been identified and thus is likely to occur through cooperation between LAT2, TAT1, and LAT4: efflux of small neutral AA in exchange for large neutral AA or aromatic AA, which is supported by the appearance of small neutral amino acids in the urine of *Lat2* knockout mice.²⁹³

LAT2 (SLC3A2-SLC7A8) shows a distinct asymmetry with regard to affinity on the two faces of the membrane. The K_M values on the outside range from 40–200 μM, while on the inside, K_M values range from 3–30 mM.²⁹⁴ The transporter accepts all neutral amino acids except proline. TAT1 (SLC16A10) shows very low affinity for its substrates, ranging from 3–7 mM.²⁹⁵ Although the transporter is related to monocarboxylate transporters, it does not cotransport protons.²⁹⁰ Apart from aromatic amino acids, TAT1 also transports the N-methylated derivatives of these amino acids and L-dopa.

LAT4 (SLC43A2) belongs to a separate transporter family than 4F2hc-LAT2. It does not require additional subunits and transports its substrates with low affinity (K_M for phenylalanine is 5 mM) on both sides of the membrane.²⁹⁶

The kidney contributes to gluconeogenesis and pH regulation. The glutamine, asparagine, and histidine transporter SNAT3 (SLC38A3) is located in the basolateral membrane but is typically expressed at low levels. During chronic acidosis, it is upregulated and glutamine imported for deamination by phosphate-activated glutaminase.²⁹⁷ The resulting glutamate is further deaminated to 2-oxoglutarate, which serves as a substrate for gluconeogenesis²⁹⁸ (see also Fig. 8.4). Ammonia is released into the urine, thereby disposing of protons. Upregulation of SNAT3 mRNA during chronic metabolic acidosis involves promoter regulation and mRNA stability.²⁹⁹ The related transporter SNAT (SLC38A5) is also expressed in the kidney and may contribute to glutamine uptake.³⁰⁰ Its subcellular localization is unknown.

There is no clear pathway for the basolateral exit of proline. Although glycine is inefficiently transported by many neutral amino acid transporters, including 4F2hc-LAT2, the latter does not accept proline. Release of taurine also remains unclear. The GABA/betaine transporter BGT1 (SLC6A12) has been identified in the basolateral membrane of renal epithelial cells but accumulates its substrates in the cytosol and therefore is not a feasible efflux pathway.

TRANSPORTERS FOR CATIONIC AMINO ACIDS AND CYSTINE

APICAL TRANSPORTERS

The presence of a transporter for cationic amino acids that is shared with cystine can be deduced from the aminoaciduria observed in cystinuria, which includes both groups of amino acids.³⁰¹ Renal cystine clearance is close to GFR in cystinuria, whereas cationic amino acids remain partially reabsorbed. Cationic amino acids and cystine are transported in the proximal tubule by the heteromeric transporter rBAT/b⁰+AT (broad neutral and cationic amino acid transporter).³⁰² Cystinuria is an autosomal recessive disorder. Homozygous (or compound heterozygous) mutations in the rBAT encoding gene SLC3A1 are classified as type A cystinuria, whereas homozygous (or compound heterozygous) mutations in the b⁰+AT encoding

gene SLC7A9 are classified as type B.³⁰³ This genetic heterogeneity can be detected in the heterozygous state, where rBAT heterozygotes do not show residual aminoaciduria, while b⁰⁺AT heterozygotes show some release of cystine and lysine into the urine. Cystinuria causes formation of kidney stones due to the low solubility of cystine.³⁰¹ The distribution of rBAT along the proximal tubule shows increasing expression toward the PST, whereas b⁰⁺AT shows the opposite trend. This suggested that rBAT may have a different partner in the PST, which was shown to be the aspartate/glutamate transporter AGT1.³⁰⁴ Initially AGT1 was thought to have a basolateral localization; however, follow-up studies using more specific antibodies demonstrated an apical localization. Functionally, AGT1 operates as a cystine transporter, exchanging cystine for glutamate. Interestingly, this suggests that cystine is transported as a neutral amino acid in the PCT and as an anionic amino acid in the PST. In b⁰⁺AT knockout mice fractional excretion of arginine reaches 80%, while only 11% of the tubular cystine load is excreted.³⁰⁵ This is consistent with the presence of additional cystine transporters, such as AGT1 and EAAT3 (see anionic amino acid transporters later). The rBAT protein is a highly glycosylated type II membrane protein that is connected by a disulfide bridge to the transporter subunit b⁰⁺AT just outside the membrane.³⁰⁶ Heterodimer formation is essential for the exit of the complex from the endoplasmic reticulum.³⁰⁷ Transport properties of rbat/b⁰⁺AT have been largely elucidated using the oocyte endogenous transporter that associates with rbat when its cRNA is expressed in this system.³⁰⁸ These experiments show that rbat/b⁰⁺AT is an obligatory antiporter that preferentially takes up cationic amino acids from the lumen in exchange for neutral amino acids. This directionality is imposed by the inside-negative cell membrane potential, and it is also confirmed by the lack of neutral amino acids in cystinuria.³⁰⁹ Affinities for cationic amino acids and cystine are approximately 100 μM ; neutral amino acids have slightly higher apparent K_M values.³¹⁰ The acidic amino acid–cystine exchanger AGT1 transports cystine with a K_M of 68 μM .³⁰⁴ In mice, the gene is only expressed in males and thus is unlikely to be essential for renal cystine reabsorption. In humans, isolated cystinuria, a rare disorder in which only cystine is elevated in urine, is caused by selected mutations in b⁰⁺AT that affect the substrate selectivity of the transport subunit.³¹¹ Mutations in AGT1 have not been identified thus far.

Clinical Relevance

Transporters for Cationic Amino Acids and Cystine Apical Transporters

Urolithiasis occurs in most cases of cystinuria. The generation of kidney stones is managed through the combination of several treatments. Tiopronin (α -mercaptopropionylglycine) is administered to form adducts with cysteine, which have a higher solubility than cystine. Potassium citrate is used to increase the pH of urine >7.5. Reduction of animal protein intake and nocturnal fluid intakes are recommended. All treatments are aimed to reduce cysteine formation and precipitation.

BASOLATERAL TRANSPORTERS

A basolateral exit pathway for cationic amino acids is defined by the disease LPI.^{256,312} LPI is caused by mutations in the

SLC7A7 light chain of the heteromeric transporter 4F2hc-SLC7A7.^{313,314} The disease is very rare, with less than 200 cases reported, mainly from the Finnish population, where a founder mutation exists. In contrast to apical transport disorders, which are often benign, LPI can be a very severe disease, although clinical symptoms vary widely.²⁵⁶ It also has a number of extrarenal pathologies, such as alveolar proteinosis and immune defects, which are not well understood. Plasma levels of cationic amino acids are reduced, which affects urea cycle function, causing the adverse reaction to protein ingestion. Urine levels of cationic amino acids are very high, particularly that of lysine. Intestinal absorption of cationic amino acids is also affected. Although a functionally redundant transporter exists (4F2hc-SLC7A6), its expression is low and cannot replace malfunction of SLC7A7.³¹⁵ Functionally, SLC7A6 and SLC7A7 are characterized as system y⁺L, indicating a transporter that exchanges neutral and cationic amino acids, which was initially discovered in placenta and erythrocytes.³¹⁶ The y⁺LAT1 (SLC7A7) transporter accepts neutral and cationic amino acids with high affinity (K_m values about 20 μM), but the affinity for neutral amino acids is two orders of magnitude lower in the absence of Na⁺.³¹⁷ As a result, the preferential mode of operation is an efflux of cationic amino acids in exchange for extracellular neutral amino acids. The y⁺LAT1 transporter is widely expressed in the basolateral membrane of the proximal tubulus.

The prevalence of exchangers for the transport of cationic and neutral amino acids sets these mechanisms apart from the paradigm set by glucose reabsorption. One possible reason could be the maintenance of cytosolic amino acid pools that are significantly higher than those observed in blood plasma and which are required for protein biosynthesis and amino acid homeostasis.³¹⁸ In the presence of basolateral facilitators, cytosolic amino acid pools would be close to those observed in the blood.

TRANSPORTERS FOR ANIONIC AMINO ACIDS

The main apical transporter for anionic amino acids in the proximal tubule is EAAT3 (named EAAC1 in rodents, SLC1A3).³¹⁹ EAAT3 transports both glutamate ($K_M = 14 \mu\text{M}$) and aspartate and shows a preference of D-aspartate over L-aspartate.³²⁰ As pointed out earlier, EAAT3 also contributes to cystine transport,³²¹ but cystine is only marginally elevated in individuals with dicarboxylic aminoaciduria.³²² The latter is a rare condition caused by mutations in EAAT3.³²² It is readily detected by highly elevated urine levels of aspartate and glutamate; excretion can reach or even exceed the filtered amounts. In addition to the kidney, EAAT3 is expressed in neurons and the intestine and allows high cytosolic accumulation due to the cotransport of 3Na⁺ and 1H⁺ and the antiport of 1K⁺.³²³ Despite expression of EAAT3 in neurons, dicarboxylic aminoaciduria is considered a benign disorder.³²⁴ However, an association of EAAT3 mutations with obsessive-compulsive disorder has been reported.³²⁵ Expression of EAAT3 is relatively low in the PCT, increases toward the PST, and is also observed in the distal parts of the tubule.³¹⁹ Nevertheless, 90% of the filtered anionic amino acid load is reabsorbed in the PCT. The transporter is regulated by osmolarity and amino acid deprivation.³²⁶ A dedicated efflux pathway for glutamate in renal epithelial cells has not been identified. In fact, there is evidence for an accumulative glutamate

transporter in the basolateral membrane, which would prevent efflux.²⁵⁹ In accordance, glutamate is intensively metabolized in epithelial cells, and efflux may be limited. Moreover, members of the SLC22 family are nonspecific anion transporters, and those located in the basolateral membrane may serve as efflux pathways for glutamate.³²⁷

STRUCTURAL INFORMATION OF AMINO ACID TRANSPORTERS

Plasma membrane amino acid transporters thus far fall into three different protein folds, namely the LeuT-fold, the Glt-fold, and the multifacilitator superfamily (MFS)-fold³²⁸ (see Table 8.4). Functionally, it has long been established that transporters must be able to adopt an inward-facing and an outward-facing conformation.^{329,330} All transporters analyzed thus far operate in an alternating access mode in which the substrate and cosubstrates (if applicable) bind on one side

of the membrane (e.g., outside). Subsequently, weak interactions between substrate and the transporter cause the transporter to enclose the substrate, resulting in the occluded conformation. The transporter transitions further into an inside-open conformation to release the substrate.³²⁸ In the case of antiporters, the energy barrier for a substrate-less translocation is too high; thus, the return to the initial conformation is only possible in a substrate-bound state. In the case of symporters and uniporters, the empty transporter can transition back through a substrate-free occluded state, back to the outside-open conformation, thereby closing the catalytic cycle. However, at least in the case of the LeuT protein fold, a pseudosubstrate in the form of a leucine side chain residing close to the substrate binding site occupies the empty binding site, thereby facilitating the transition back to the outside conformation.³³¹

For the LeuT-fold, high-resolution structures have been identified for all but one stage of the transport cycle (Fig.

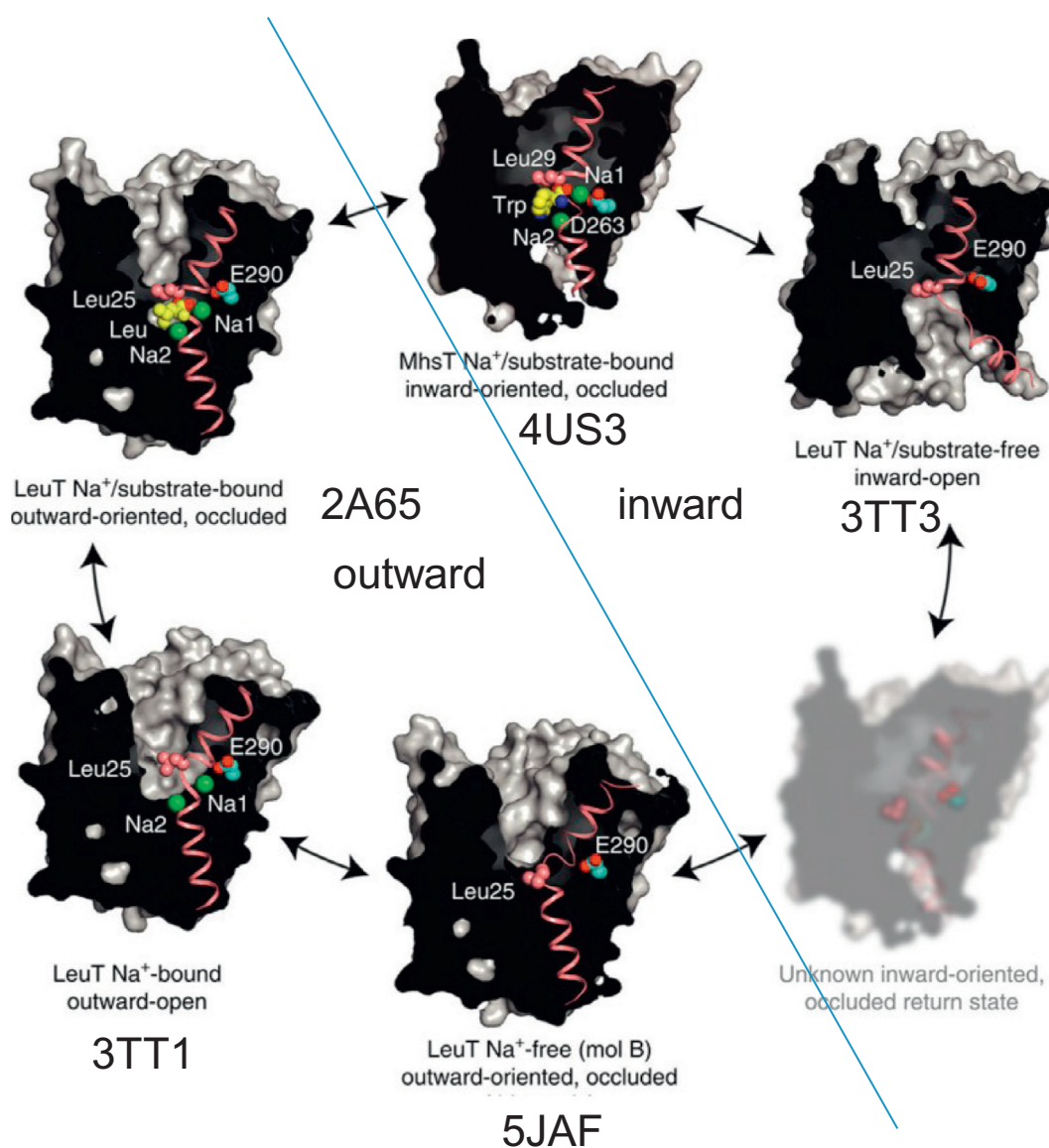


Fig. 8.14 Transport cycle of LeuT-fold proteins. TM1 helix and pseudosubstrate Leu25 (Leu29^{MhsT}) are shown in pink, the Na⁺ ions are shown as green spheres, and Glu290 (or Asp263^{MhsT}) as cyan, and substrate at the binding site as yellow spheres. (Reused from Malinauskaitė L, Said S, Sahin C, et al. A conserved leucine occupies the empty substrate site of LeuT in the Na(+)-free return state. *Nat Commun.* 2016;7:11673, under creative common license.)

8.14). The first structure of LeuT revealed an internal symmetry, whereby helix 1-5 can be superimposed onto helix 1-6 in a twofold rotation.³³² This internal symmetry provides the structural basis for the adoption of an inward- and outward-facing conformation.³³³ Further investigation of this protein fold revealed that part of the protein remains rather immobile—an arrangement of helices 3,5 and 7,8 named the hash (#)—whereas others move significantly—the bundle—composed of helices 1,2 and 6,7.³³³ The hydrogen bond network is interrupted in the center of helices 1 and 6, allowing the helix to bend and to use these hydrogen bonds to interact with substrates and cotransported ions. A key feature in most transporters with a LeuT-fold is a motion of the bundle relative to the hash or vice versa, thereby alternatively closing and opening alternating sites of the transporter.³²⁸ Another common feature is the presence of one or more aromatic amino acids adjacent to the substrate binding site, which insulate the substrate during the transition to the occluded state. These residues are referred to as “thin” gates in contrast to subsequent enclosure of the substrate by more substantial parts of the protein (thick gates).

The transport process in Na⁺-driven glutamate transporters is quite different, as it involves an elevator-like movement of the transport domain relative to a rigid scaffold domain.³³⁴ Glt_{ph} is composed of eight TM segments and two helical hairpins (HP1 and HP2).³³⁵ The six N-terminal helices form a large bracket that holds TM7, TM8, HP1, and HP2. Two large loops, one connecting TM2 and TM3 and the other TM5 and TM6, serve as the hinge to allow elevator-like movement of the substrate binding domain.³³⁶ Reminiscent of similar features in LeuT, the structure and ions are bound by unwound regions of TM7, TM8, HP1, and HP2. Alternating access is mediated by opening of helical HPs on the respective side of the membrane. Glutamate transporters form trimers, which are likely to be important to stabilize the scaffold against the movement of the elevator domain.

In the MFS fold, the protein has a symmetry axis perpendicular to the membrane between helices 1–6 and 7–12.^{337,338} During the transport cycle, the two halves of the protein move against each other. This rocker-switch mode needs to be modified to accommodate initial occlusion of the substrate by the bending of the outer ends of helices 1 and 7 on the outside or 4 and 10 on the inside. Binding of amino acids remains unclear due to the lack of high-resolution structures of amino acid-translocating members.

As pointed out earlier, several renal amino acid transporters form heterodimers with ancillary proteins—for instance, SLC7 transporters with 4F2hc and rBAT and SLC6 transporter B⁰AT1 with collectrin. The main role of ancillary subunits is to facilitate the exit of the complex from the endoplasmic reticulum.³⁰⁷ Collectrin, in addition, is also essential for the catalytic activity of the transporter.²⁸⁴ The structure of the ectodomains of transporter ancillary proteins is known for 4F2hc,³³⁹ but the precise heteromeric arrangement is only partially understood.³⁴⁰ The structure of the 4F2hc ectodomain, which is related to bacterial α -amylases, is composed of two subdomains, namely domain A, a triphosphate isomerase barrel domain ($\alpha\beta$)₈, and domain C, made up of eight antiparallel β -strands.³³⁹ However, neither 4F2hc nor rBAT has glycosidase activity and they both lack conservation of critical catalytic residues. The glycosylation of the ancillary proteins could be important to help with quality control in the endoplasmic reticulum. In addition to its role in amino

acid transport, collectrin is thought to be involved in kidney development and vesicle exocytosis, but these functions are mechanistically less well understood.³⁴¹ The 4F2 heavy chain has additional functions in integrin signaling, which are well understood in cancer cells but not in epithelial cells.^{266,306,342}



Complete reference list available at ExpertConsult.com.

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BOARD REVIEW QUESTION

1. What statement about the sodium glucose cotransporter SGLT2 in the kidney is incorrect?
 - a. SGLT2 reabsorbs >90% of the filtered glucose in an euglycemic individual with normal kidney function
 - b. SGLT2 is expressed in the early proximal tubule
 - c. SGLT2 inhibition enhances GFR in short term
 - d. SGLT2 inhibition excretes only ~50% of the filtered glucose in an euglycemic individual with normal kidney function
 - e. SGLT2 inhibition is uricosuric

Answer: c

Rationale: SGLT2 is expressed in the brush border of the early proximal tubule. SGLT2 reabsorbs >90% of the filtered

glucose in an euglycemic individual with normal kidney function. SGLT2 inhibition excretes only ~50% of the filtered glucose in an euglycemic individual with normal kidney function, because SGLT1 in the downstream later part of the proximal tubule can compensate when more glucose is delivered to that site. SGLT2 inhibition is uricosuric and can lower blood urate levels. SGLT2 inhibition reduces GFR in short term through the tubuloglomerular feedback mechanism and by increasing tubular back pressure. SGLT2 inhibition can preserve or maintain higher GFR in the long term in type 2 diabetic patients.